

Research Papers

Bayesian multi-agent deep reinforcement learning approach for improving cyber-secure load-to-grid services and market participation in storage-rich microgrids

Reza Sepehrzad ^{a,*}, Milad Moafi ^b, Nima Khosravi ^c, Ahmed al Durra ^d, Mahdiah S. Sadabadi ^e

^a Department of Electrical and Computer Engineering, University of Canterbury, Christchurch, New Zealand

^b Department of Power and Control Engineering, University of Tabriz, Iran

^c Centre of Research Impact and Outcome, Chitkara University Institute of Engineering and Technology, Chitkara University, Rajpura, 140401, Punjab, India

^d Advanced Power & Energy Center, EECS Department, Khalifa University, Abu Dhabi, United Arab Emirates

^e Department of Electrical and Electronic Engineering, University of Manchester, Manchester, United Kingdom

ARTICLE INFO

Keywords:

Microgrid cybersecurity
Multi-agent deep reinforcement learning
Load-to-grid services
Cyber-attack detection
Secure energy market operations

ABSTRACT

The growing dependence on digital communication and machine learning within microgrid (MG) systems has brought new levels of flexibility and automation, but it has also exposed these environments to increasingly coordinated cyber-attacks. This study introduces a defense strategy designed to protect trained control policies during both offline learning and real-time operation by using a Bayesian multi-agent deep reinforcement learning framework. The approach defines a safety region that continuously evaluates policy integrity with respect to load-to-grid (L2G) services and the microgrid's operational goals. In this work, L2G strategies are modeled around energy storage systems (ESS), electric vehicles (EVs), static loads, and renewable energy resources (RER), reflecting the practical structure of modern MGs and their participation in demand response and market-driven energy transactions. To identify harmful data injections and capture the evolving nature of hybrid cyber intrusions, the method combines Gaussian and Bernoulli-based modeling with Euclidean distance measures, enabling accurate detection of abnormal variable behavior. Extensive tests across multiple operating scenarios demonstrate the effectiveness of the framework. The experimental results show an 11.46% reduction in operational costs and prevent roughly \$ 6717 in potential attacker gains, underscoring the method's capacity to secure L2G interactions while supporting resilient and economically efficient microgrid operation.

1. Introduction

1.1. Background and motivation

The concept of microgrids (MG) introduced new distribution systems of energy to address issues in the power sector. However, urban community growth into smart cities and increased utilization of renewable energy resources (RER) in modern distribution systems of energy challenge the evaluation of stability, security, and reliability in power systems [1]. Security of power systems integrated with MGs relies on evaluating control algorithms to improve transition stability through large disturbances and achieve stability. Destabilization can lead to the collapse of power systems in a matter of seconds [2]. In this situation, the distribution system operator (DSO) cannot control the transition stability. Therefore, computer-based control approaches without

manual DSO interaction can be relied upon to achieve transition stability [3]. The components of MGs such as power electronics, renewable and non-renewable distributed energy resources, and multi-scale, nonlinear, and stochastic energy storage systems (ESS) have made MGs highly random and uncertain, resulting in the fundamental transformation of control strategies [4].

The development of MGs based on smart grid requirements has significantly benefited from advances in communication systems such as 5G and 6G, which enable information exchange and play a crucial role in algorithm and control policy development [5]. Thanks to the technological innovation in terms of communication infrastructure and energy market structures, end-users have come forward as active players in the energy market. Designing energy management algorithms, i.e., demand response (DR) programs, has also facilitated the deployment of load-to-grid (L2G) services [6]. By offering a two-way medium for energy

* Corresponding author.

E-mail address: Sepehrzad.reza@gmail.com (R. Sepehrzad).

<https://doi.org/10.1016/j.est.2026.122422>

Received 13 November 2025; Received in revised form 12 April 2026; Accepted 23 April 2026

Available online 30 April 2026

2352-152X/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

trading, the energy market can play a critical role in consumer involvement in L2G services. However, cyber-attacks over communication systems represent a significant challenge for the evolution of MGs and L2G services [7]. The communication system is vulnerable to hybrid cyber-attacks like false data injection (FDI) and denial of service (DoS), which can cause confusion, instability, and reduced reliability. These cyber-attacks, infiltrating different layers of MGs, can result in irreparable damage to the market structure and power exchange. Developing a control mechanism to defend against hybrid cyber-attacks is a critical challenge for the MG structure which plays an effective role in the development of this structure [8]. In addition, the vulnerability of renewable-integrated MGs to post-fault cascading contingencies and the growing threat of coordinated cyber-physical attacks that jointly manipulate physical and cyber layers further confirm the need for resilient and secure control mechanisms in modern MG operation [9,10].

1.2. Problem statement

Cyber-disruptive attack vectors in microgrid structures create major difficulties for detection and control frameworks because their dynamic behavior increases both computational complexity and processing time. A critical issue that has received limited attention in earlier studies is the vulnerability of machine learning (ML)-based policies and control algorithms to cyberattacks. When communication infrastructures are compromised, both the offline training stage and the online execution stage of ML methods may be affected, which can degrade or even invalidate the learned control policies. This gap remains insufficiently examined in the existing literature, despite its potential to cause severe operational consequences.

1.3. Literature survey

This section reviews the performance improvement of the microgrid structure through participation in the energy market, L2G services, and the capabilities of reinforcement learning (RL) and deep reinforcement learning (DRL) for mitigating hybrid cyberattack threats (FDI/DoS).

1.3.1. MG market structure and hybrid cyber-attacks

The energy market provides an effective framework for implementing L2G services through demand response programs and active end-user participation in optimal energy allocation. However, this market remains exposed to cyberattacks that can undermine its reliability and, as a result, slow the advancement of microgrid development [11]. With the increasing adoption of electric vehicles (EVs), charging demand in high-traffic or poorly served areas often exceeds the capacity of fixed charging stations (FCSs). To address this issue, Ref. [11] introduces a peer-to-peer (P2P) energy trading mechanism between EVs and mobile charging stations (MCSs), allowing EVs to obtain energy from MCSs instead of depending exclusively on FCSs. The study develops an auction-based incentive mechanism in which MCSs act as auctioneers and EVs as bidders, with the aim of ensuring fairness, valid price formation, and fraud prevention. The proposed framework satisfies key economic and computational properties, including truthfulness, individual rationality, budget balance, and computational efficiency. Despite these advantages, a broader concern remains the possibility of bypassing control mechanisms or injecting false data, which can compromise the integrity of the energy trading platform. Ref. [12] examines the effect of cyberattacks on the power market, with particular attention to outages caused by sabotage in a three-bus wind power system. The proposed approach formulates a post-attack cost minimization problem as a mixed-integer linear programming model solved using CPLEX. In addition, the output data generated in GAMS are further processed in Python for subsequent analysis through histogram-based evaluation and machine learning techniques. The most significant result is that cyberattacks redefine the maximum point of load and,

among the machine learning models experimented, a two-layer neural network possesses the greatest predictability and separability in identifying and separating attacks, with the ultimate advantage of improving system security. However, the inherent shortcoming of such a method still lies in its incapability to deal with dynamic data and the exponential increase in computing load and processing time.

Ref. [13] is motivated by the increasing concurrent risk of extreme weather events and cyberattacks, both of which can severely disrupt critical infrastructure systems. To analyze these threats, the study develops a two-stage framework that captures disruptions in electric power systems and their broader economic consequences under three conditions: Heatwave, Cyber-attack, and Compound, where both events occur simultaneously. The proposed methodology integrates a bilevel optimization model to represent attacker behavior, an optimal power flow model to assess electricity-system disruption, and a Computable General Equilibrium model to estimate economy-wide impacts. The results show that a heatwave alone is unlikely to produce a feasible power deficit scenario, whereas a cyberattack leads to 4% unserved electric load in Long Island, and the compound scenario increases this value to 12%, affecting nearly 198,000 customers. In addition, government enterprise activity declines by 37% in the compound case, indicating substantial societal and operational consequences. Despite these contributions, the approach still faces an important limitation, as its control policies remain vulnerable to false data injection attacks and its ability to process dynamic data in real time is relatively limited. Ref. [14] investigates adversarial attacks against electricity markets with the motivation to identify and prevent malicious operations due to the fact that the power network complexity does not allow it. The attack strategy employs linear optimization and statistical learning for developing surreptitious attacks that, while indistinguishable from the optimal solutions, are stealthy. Through framing market dynamics as an EPEC problem and its resolution via Fisher-Burmeister function and strong duality theorem, the research investigates economic, policy, and societal implications of these attacks. The critical observations reveal that microgrids can change utilization by up to 12% nobody would be any wiser about, or between 14% to 22% variations based on surveillance capabilities. Moreover, while more solar generation can counterbalance the impact of assaults, it also brings new vulnerabilities. Nevertheless, one notable drawback of this approach is that it has a low processing rate and dynamic response.

1.3.2. L2G service structure and cyber-attacks

The energy market and DR programs enable end-users to participate in L2G services, offering benefits such as voltage and frequency control, reduced peak load, improved stability, and lower service/maintenance costs. DSOs can use L2G services to enhance the stability and normal operation of the MG. Cyber-attacks with destructive objectives can impact L2G services and have serious consequences for the stability and reliability of MGs. Hence, control approaches and policies against cyber-attacks are crucial for the normal operation of MGs [15]. The motivation of Ref. [16] is the vulnerability of smart grid demand-response schemes to cyber-attacks that inject spurious data on consumption and generation, threatening grid stability. To mitigate this threat, the research presents an online detection technique using convolutional neural networks (CNNs) to identify and nullify such attacks. The experimental results indicate that power grid distribution networks are most susceptible to such cyber-attacks, but the CNN-based detector proposed here functions effectively in detecting and removing fixed change rate attack events. However, the main weakness for this approach is that it fails to handle dynamic data properly, thus having lower reliability when dealing with incidents involving data injection and price manipulation, particularly where there is an attack with a varying change rate. Ref. [17] addresses the challenge posed by RES uncertainty and cyber-attacks in MGs for load frequency control (LFC). As measures against them, the paper investigates the use of hydrogen as an energy reservoir through the combined system of fuel cell and electrolysis systems in mitigating

the power fluctuation. An integrated MATLAB/Simulink model of a microgrid with the cyber-attack mechanism is simulated, and an optimal dual-input interval type 2 fuzzy logic controller - fractional order tilt-integral-derivative cascade controller (DIT2FLC-FOTID) is introduced for LFC regulation. The proposed controller minimizes the frequency response significantly by performing better than the controllers by over 67% when compared to the Fuzzy TID controller and over 87% compared to the FOTID controller. The process itself, though, carries a significant disadvantage of having a low processing rate and high dynamic response, leading to increased processing time in operations.

Ref [18] addresses the problem of optimally scheduling multi-energy flow in integrated energy system (IES) in the face of uncertainties of RES, loads, and the larger threat of cyber-attacks. To solve the problems, the authors propose a state-adversarial DRL framework, which integrates a model-free resilience scheduling scheme based on an integrated demand response (IDR) program and the SA-MDP model. This strategy uses the SA-SAC algorithm to develop a stronger framework against cyber-attacks and schedule more optimally. Results show that the method can react adequately to uncertainties provoked by RES and loads, counteract effects of cyber-attacks, and improve economic operation by 10% compared to the native soft actor critic (SAC) algorithm. The major drawback of the method is the very high computing burden and computational time needed in implementing the DRL model complexity. Ref. [19] proposes an intelligent data-driven hierarchical energy management policy aimed at enhancing the roles of renewable energy resources, energy storage systems, and incentive-based demand response programs within a multi-energy integrated system (MIES). The policy is designed to minimize operating cost, environmental emissions, risk-related cost, and the effects of cyberattacks such as false data injection, and it is implemented through multi-agent deep reinforcement learning (MADRL) formulated on Markov decision processes. The reported results indicate notable reductions of 19.51% in operating cost, 19.69% in risk cost, 24% in cybersecurity-related cost, and 20.24% in pollution. The framework also targets smart city applications by offering fast response, high accuracy, and relatively low computational burden. Despite these advantages, the method still faces important challenges, particularly the deterioration of control policies under false data injection attacks and reduced reliability when deployed in systems with large-scale data volumes.

1.3.3. Advanced approaches and cyber-attacks

Due to advances in ML algorithms like RL and DRL, control approaches and policies for MGs have evolved significantly. ML-based algorithms have improved data evaluation and increased the accuracy of control policies for complex characteristics of MGs. RL and DRL methods have proven to be highly effective in handling dynamic and time-varying data, such as hybrid cyber-attacks [20]. Ref [20] addresses the increasingly urgent issue of false data injection attacks (FDIAs) in smart grids, which would jeopardize their stability and security. The resolution proposes a DRL detection algorithm with the backup of an attention mechanism to better extract state features and stress on important information. With the differential attention, the system can recognize significant features and inhibit meaningless features, and hence optimize detection efficiency and accuracy. The major findings are that the proposed method is efficient in detecting attacks, with improved detection rates compared to similar methods when used on IEEE-14 and IEEE-118 bus systems. The major problem, however, is the potential disruption of control policies by FDIAs and the reduced reliability of the method in large data environments, where the system may fail to function. Ref. [21] is motivated by security concerns in internet of energy (IoE)-based smart grids, particularly their susceptibility to false data injection attacks that can disrupt power scheduling, routing strategies, and overall grid operation. To address this issue, the study develops a reinforcement learning-based attack generator to emulate network conditions and produce both known and unknown attack patterns. It then applies a multilayer defense framework based on snapshot

ensemble deep neural networks and deep autoencoder networks for attack detection. The reported results indicate that the method can detect passive and active attacks effectively, achieving an accuracy of 98.82% with 97.42% false positives. Despite these results, the approach still suffers from major limitations, including the degradation of control policies under improper data injection, relatively low operational processing speed, and high computational overhead. Ref. [22] addresses the vulnerability of electric vehicle charging stations (EVCSs) to cyber uncertainties, such as communication delay and false data injection attacks, by proposing a techno-economic evaluation framework based on the Takagi-Sugeno-Kang fuzzy system and multi-agent deep reinforcement learning (TSKFS&MADRL). This method is designed to identify and compensate for cyber uncertainties, thereby improving dynamic response and strengthening EVCS resilience. The findings show that the proposed approach reduces operating cost by 7.33% compared with RL and by 12.15% compared with CNN, while also decreasing detection time by 40%. However, its main limitation is the weak suitability for systems with large-scale data, where the processing speed declines significantly. Table 1 provides a comprehensive comparison between the proposed approach and related studies.

1.4. Main contribution and highlights

The rapid expansion of communication technologies, digital energy platforms, and ML-enabled control has accelerated the development of microgrids in line with smart-city requirements. At the same time, this increasing digitalization has widened the attack surface of microgrid operation and exposed critical control processes to more complex hybrid cyber threats (FDI/DoS). Such threats do not only disturb communication signals and operational data, but can also degrade the integrity of trained control policies during both online execution and real-time decision making. To address this challenge, this study proposes a Bayesian multi-agent deep reinforcement learning (BMADRL) framework designed to enhance the cyber-resilience of microgrids while preserving the operational effectiveness of L2G services. The proposed framework establishes a secure evaluation space for continuously validating policy behavior against the operational objectives of the microgrid and the dynamic requirements of L2G participation.

Within this structure, the flexibility of L2G services is modeled through the coordinated participation of static loads, EVs, ESSs, and RERs, since these components directly shape the real-time balancing capability between demand and supply. Their dynamic behavior is explicitly linked to the resilience and stability of the microgrid under uncertain and attack-aware operating conditions. In addition, the framework incorporates both market and communication layers. The market layer is represented through wholesale and retail structures together with incentive-based demand response (IBDR) programs, while the communication layer is formulated using graph-theoretic principles to describe the signal-exchange structure among microgrid entities. These two layers are especially important because they form both the operational backbone of L2G services and the main entry points for cyber-attacks.

The detection of hybrid cyber-attack vectors is inherently challenging because malicious disturbances may appear as both continuous and discrete anomalies in dynamic operating data. To address this problem, the proposed framework combines Gaussian distribution modeling, Bernoulli-based anomaly interpretation, Euclidean-distance-based safe-region assessment, and weighted least-squares residual analysis in order to detect intrusive and malicious data more effectively. A spherical safe region is constructed to distinguish normal operational data from cyber-compromised observations. Data remaining inside this region are treated as secure and consistent with expected system behavior, whereas data outside the admissible boundary are classified as suspicious or malicious. The Bayesian deep reinforcement learning (BDRL) component further strengthens this structure by validating the reliability of control actions against observation-supported belief

Table 1

Comparison of the suggested method with some related literature.

References	Stability assessment	Market assessment	Computational burden and time assessment	Decision-making errors assessment	Protection of trained policies	ML methods
Ref [11]	*	*	—	—	—	—
Ref [12]	*	*	—	—	—	*
Ref [13]	*	*	—	—	—	—
Ref [14]	*	*	*	—	—	—
Ref [15]	*	—	*	*	—	*
Ref [16]	*	—	*	*	—	—
Ref [17]	*	—	—	*	—	—
Ref [18]	—	*	*	—	—	*
Ref [19]	*	*	*	—	—	*
Ref [20]	*	*	*	—	—	*
Ref [21]	*	—	—	*	—	*
Ref [22]	*	*	*	—	—	*
Proposed Method	*	*	*	*	*	*

updates, thereby reducing the destructive effect of hybrid cyber-attacks on trained policies and improving the robustness of real-time execution. Experimental evaluation under multiple operating scenarios confirms that the proposed framework provides strong detection capability, lower computational burden, fast response, and resilient policy preservation under cyberattack conditions.

The main contributions of this study are summarized as follows:

1. A cyber-secure BMADRL framework is developed to improve the resilience and stability of microgrids under hybrid cyber-attacks while preserving L2G service functionality through the integrated modeling of market interaction, incentive-based demand response, and secure communication processes.
2. A robust policy-protection and action-validation mechanism is introduced by combining MADRL-based control learning with Bayesian belief-supported verification, so that trained control policies can be protected against degradation during both offline training and real-time operation.
3. A hybrid cyber-attack detection and secure evaluation mechanism is formulated through Gaussian and Bernoulli probabilistic modeling, Euclidean-distance-based safe-region analysis, and residual-based dynamic data assessment, enabling computationally efficient identification of malicious deviations and supporting accurate, fast, and resilient microgrid operation under adversarial conditions.

The rest of the paper is arranged as: MG structures are presented in [section 2](#) relying on the L2G service model and hybrid cyber-attack detection algorithms. [Section 3](#) establishes BMADRL. The experimental results and operation scenarios are presented in [section 4](#). The conclusion is given in [section 5](#).

2. Problem formulation: microgrid under hybrid cyber attacks

The structure of the market and communication system layers is first described, followed by the configuration of L2G services under incentive-based demand response programs and the associated power exchange mechanism between buyers and sellers. Within this framework, ESSs, EVs, static loads, and renewable energy resources constitute the central elements of the trading process, because their capacity to supply or absorb power directly influences system balance, operational flexibility, and the overall effectiveness of L2G service provision.

2.1. Market structure

The market structure is configured based on the wholesale market, the retail market, and the end-user model.

2.1.1. Wholesale and retail market

In the wholesale market, the sellers set is indicated by $Se = \{se_1, \dots, se_n\}$ and the buyers set is indicated by $Bu = \{bu_1, \dots, bu_m\}$. For each seller (se), the sales quantity (q_t^{se}) along with the minimum price for sale in the time interval t is represented by the $f_t^{se}(q_t^{se})$. Also, for each buyer (bu), the purchase quantity along with the maximum price for purchase in the interval t is shown with the $f_t^{bu}(q_t^{bu})$ [23]. In the wholesale market, if MG is considered without losses and congestion, then the problem of maximizing social welfare is described according to the following equations.

$$\max_{\substack{q_t^{se_1}, \dots, q_t^{se_n} \\ q_t^{bu_1}, \dots, q_t^{bu_m}}} \sum_{b \in B} \int_0^{q_t^{bu}} f_t^{bu}(q) dq - \sum_{g \in G} \int_0^{q_t^{se}} f_t^{se}(q) dq, \quad (1)$$

Subject to:

$$\sum_{bu \in Bu} q_t^{bu} - \sum_{se \in Se} q_t^{se} = 0 \quad \left| \quad q_t = \sum_{bu \in Bu} q_t^{bu} = \sum_{se \in Se} q_t^{se} \right. \leftrightarrow \mathbb{F}_t, \quad (2)$$

$$(q_t^{se_1}, \dots, q_t^{se_n}, q_t^{bu_1}, \dots, q_t^{bu_m}) \in Q_t \quad (3)$$

Eq. (2) shows the power balance, $\mathbb{F}_t$ is the dual variable related to the power balance equation and represents the uniform price, Q_t is a possible set of decision variables based on the market clearing price and the quantities of energy sales and purchases in the previous period. Considering the IBDR program implementation based on L2G services and the simplicity of the analysis, it is possible to justify the equality condition of purchase and sale quantities for this problem, which is shown in (2) with the $q_t = \sum_{bu \in Bu} q_t^{bu} = \sum_{se \in Se} q_t^{se}$. The structure of the wholesale market, based on the IBDR program signals, describes a uniform market in each period [23]. The quantity offered in the uniform market by the customer and the seller can be justified by considering the competitive structure according to the following equation.

$$q_t^{bu,se} = \operatorname{argmax}_{q_t^{bu,se}} q_t^{bu,se} \quad \left| \quad f_t^{bu,se}(q_t^{bu,se}) \geq \mathbb{F}_t \right. \quad (4)$$

In the retail market, energy distribution entities (EDEs) have the task of distributing the power exchanged with the wholesale market according to the supply and demand indicators in the IBDR program and the L2G services configuration [24,25]. In the retail market structure, the quantity of exchanged energy is considered as $q_t^{Ex} = q_t^{bu,se}$ and the retail market clearing price is considered as p_t^{Clear} . The set of EDEs in the retail market structure is represented by $En = \{en_1, \dots, en_i\}$ and the subscribers set is shown by $l_t = \sum_{en \in En} l_t^{en} = f(l_t^{en}, p_t^{Clear})$, and each EDE is

considered as an aggregator. To maximize social welfare in the retail market structure, the following equation can be configured.

$$\max_{p_t^{clear}, p_{t+1}^{clear}, \dots, p_t^{clear}, p_t^{clear}} E_{OP} \left[\sum_{t=1}^{T=24} I_t \left((p_t^{clear} - \mathbb{F}_t) l_t - D_t (l_t - q_t^{bu,se}) \right) \right] \quad (5)$$

where E_{OP} is the expected operation based on IBDR program and L2G services. $I_t \in [0, 1)$ is the upward coefficient that increases the future profit, D_t is also a non-negative scalar function that describes the profit discount factor caused by the lack of commitment to participate in the programs provided by the market, $\bar{p}_t^{clear}, \underline{p}_t^{clear}$ represent the upper and lower limits of the retail market clearing price. According to (6), the behavior pattern of each EDE in energy distribution is also described based on the normal probability distribution function.

$$\begin{cases} \max_{l_t^{en} \in L_t^{en}} \beta^{en} (e_t^{en}, l_t^{en}) - p_t^{clear} l_t^{en} \\ e_{t+1}^{en} = e_t^{en} + \underbrace{\eta_t^{en} (e_t^{en} - l_t^{en}) + \xi_t^{en}}_{\text{penalty factor}} \Rightarrow f(e_t^{en}) = \frac{e^{-\left[\frac{(e_t^{en} - \mu)^2}{2\sigma^2} \right]}}{\sqrt{2\pi\sigma^2}} \end{cases} \quad (6)$$

where e_t^{en} is the energy required by each entity in the retail market, β^{en} is the profit function of each EDE, $\eta_t^{en} (e_t^{en} - l_t^{en}) + \xi_t^{en}$ is the penalty factor and contains $\eta_t^{en} \in [0.1]$ which is the delay rate and represents the percentage of unsupplied energy demand, and ξ_t^{en} represents a variable which sets the rate of power exchanged changes in each time slot for each entity. L_t^{en} also describes the feasible set of subscriber-side power. Also $f(e_t^{en})$ describes the load probability behavior in each EDE with the normal probability distribution function. σ^2, μ show the variance and mean of the normal probability distribution function. According to (7), the penalty factor rate can be minimized according to the consumption and generation commitment.

$$\underbrace{\eta_t^{en} (e_t^{en} - l_t^{en}) + \xi_t^{en}}_{\text{penalty factor}} \Rightarrow \min \left(\sum_{k=1}^K \rho_G \cdot \Delta g_{seller}^k + \sum_{k=1}^K \rho_L \cdot \Delta l_{buyer}^k \right) \quad (7)$$

where ρ_G, ρ_L are the generation and consumption commitment coefficients, $\Delta g_{seller}^k, \Delta l_{buyer}^k$ are the generation and consumption power changes in the market execution rounds (k).

2.2. Communication system configuration

The descriptive theory of the communication system in the MG structure is configured based on the bidirectional graph theory $G = (V, E, A)$, which includes the set of nodes, the set of edges, and the adjacency matrix. V is the set of nodes between different components in MG and they consist of $V\{v_i : i \in \mathbb{N}\}$. E is the set of edges that consist of $E \subseteq V \times V$. $(v_i, v_j) \in E$ is described when there is information and data exchange between nodes i and j in the communication system. A is the adjacency matrix consisting of $A \in \mathbb{R}^{N \times N}$ and $a_{ij} = a_{ji}, a_{ii} = 0$. If there is information exchange between nodes i and j , then $a_{ij} > 0$, otherwise $a_{ij} = 0$ [5]. Also, according to the adjacency matrix, the neighbors' set of MG components can be described according to (8).

$$N_i = \{v_j \in V : (v_i, v_j) \in E\} \quad (8)$$

The adjacency Laplace matrix of graph G is:

$$\mathcal{L} = [\mathcal{L}_{ij}] \in \mathbb{R}^{N \times N} \mid \mathcal{L}_{ij} = \sum_{j \neq i} a_{ij} \quad (9)$$

Hackers can expose different structures of MGs by uploading false

information to the main signals on the communication system. It is possible to monitor the changes loaded on the original data exchange by describing the symmetric information exchange matrix based on the adjacency matrix of graph G and the diagonal matrix of graph G .

$$\begin{cases} \mathcal{L}^{\text{Symmetric}} = \mathcal{L}^{\text{Adjacency}} + \mathcal{L}^{\text{Diagonal}} \\ \mathcal{L}^{\text{Diagonal}} = \text{diag}[a_{i0}] \begin{cases} \text{if receive information from node } i \\ \text{otherwise, } a_{i0} = 0 \end{cases} \end{cases} \quad (10)$$

where $\mathcal{L}^{\text{Symmetric}}$ shows the symmetric information exchange matrix, $\mathcal{L}^{\text{Adjacency}}$ is the adjacency matrix of graph G and $\mathcal{L}^{\text{Diagonal}}$ is the diagonal matrix of graph G .

2.3. L2G services configuration

The L2G service configuration equations are formulated on the basis of energy market interactions and incentive-based demand response programs, with the objective of establishing a competitive and secure environment for subscribers. In this study, the optimization of L2G services is modeled using Multi-Task Learning within a multi-objective optimization framework. Further details are provided in [26].

2.3.1. L2G service objective functions

According to Eq. (11), the objective function of L2G services is described, which include MGs economic operation, risk assessment, and implementation of the L2G services structure for regulating MG's voltage and frequency.

$$O.F_{MG} = \min \sum_{t=1}^{T=24} f_{OP,t}^{MG} + f_{Risk,t}^{MG} + f_{L2G,t}^{MG} \quad (11)$$

$$f_{OP,t}^{MG} = \sum_{t=1}^{T=24} \left(\underbrace{\sum_{t=1}^{T=24} C_t^{\text{grid}}}_{\text{main grid Operation Cost}} + \underbrace{\sum_{t=1}^{T=24} C_t^{\text{RER}}}_{\text{RER Operation Cost}} + \underbrace{\sum_{t=1}^{T=24} C_t^{\text{ESS}}}_{\text{ESS Operation Cost}} + \underbrace{\sum_{t=1}^{T=24} C_t^{\text{IBDR}}}_{\text{IBDR Operation Cost}} + \underbrace{\sum_{t=1}^{T=24} C_t^{\text{Load}}}_{\text{Load Operation Cost}} - \underbrace{\sum_{t=1}^{T=24} C_t^{\text{EV}}}_{\text{EV Operation Cost}} \right) \quad (12)$$

$$C_t^{\text{grid}} = \sum_{t=1}^{T=24} (\delta_{\text{grid}}^{\text{IBDR}}) \cdot [(\Lambda_t^{\text{grid,se}} \cdot P_t^{\text{grid,se}}) - (\Lambda_t^{\text{grid,bu}} \cdot P_t^{\text{grid,bu}})] \quad (12-1)$$

$$C_t^{\text{RER}} = \sum_{t=1}^{T=24} (\delta_{\text{RER}}^{\text{IBDR}}) \cdot (\Lambda_t^{\text{WT}} \cdot P_t^{\text{WT}} + \Lambda_t^{\text{PV}} \cdot P_t^{\text{PV}}) \quad (12-2)$$

$$C_t^{\text{ESS}} = \sum_{t=1}^{T=24} (\delta_{\text{ESS}}^{\text{IBDR}}) \cdot (\Lambda_t^{\text{ESS,se}} \cdot P_t^{\text{ESS,se}} - \Lambda_t^{\text{ESS,bu}} \cdot P_t^{\text{ESS,bu}}) \quad (12-3)$$

$$C_t^{\text{IBDR}} = \sum_{t=1}^{T=24} (\Lambda_t^{\text{IBDR}} \cdot [P_t^{\text{Up}} - P_t^{\text{Down}}]) \quad (12-4)$$

$$C_t^{\text{Load}} = \sum_{t=1}^{T=24} (\delta_{\text{Load}}^{\text{IBDR}}) \cdot \left[\left(\Lambda_t^{\text{Load}} \cdot P_t^{\text{Load}} \pm \sum_{t=1}^{T=24} (\Lambda_t^{\text{Load}} \cdot \Delta P_t^{\text{Load}}) \right) \right] \quad (12-5)$$

$$C_t^{\text{EV}} = \sum_{n=1}^N N_{EV} \cdot \sum_{t=1}^{T=24} (\delta_{EV}^{\text{IBDR}}) \cdot (\Lambda_t^{\text{EV,se}} \cdot P_t^{\text{EV,se}} - \Lambda_t^{\text{EV,bu}} \cdot P_t^{\text{EV,bu}}) \quad (12-6)$$

$$\pm P_t^{\text{grid}} \pm P_t^{\text{ESS}} \pm P_t^{\text{EV}} \pm P_t^{\text{Load}} + P_t^{\text{RER}} = 0 \quad (12-7)$$

$$P_{\min}^{grid} \leq P_t^{grid} \leq P_{\max}^{grid} \quad (12-8)$$

$$P_{\min}^{RER} \leq P_t^{RER} \leq P_{\max}^{RER} \quad (12-9)$$

$$P_{\min}^{ESS} \leq P_t^{ESS} \leq P_{\max}^{ESS} \quad (12-10)$$

$$P_{\min}^{EV} \leq P_t^{EV} \leq P_{\max}^{EV} \quad (12-11)$$

$$P_{\min}^{Load} \leq P_t^{Load} \leq P_{\max}^{Load} \quad (12-12)$$

According to (11), the objective function of the MG structure is presented such that, $f_{OP,t}^{MG}$ represents the MGs' economic operation, $f_{Risk,t}^{MG}$ represents the risk assessment, and $f_{L2G,t}^{MG}$ represents the configuration of L2G services. According to (12), the set of components affecting the objective function of the first level such as operating costs of the main grid, RERs, ESSs, IBDR program implementation, EV units and loads are categorized. Eqs. (12-1) to (12-6) describe the components affecting the operation costs of the first level objective function, so that, $\Lambda_t^{grid,se}$, $\Lambda_t^{grid,bu}$ represent the operation cost and energy exchange between the main grid and MG, Λ_t^{WT} , Λ_t^{PV} represent the RER operation costs, $\Lambda_t^{ESS,se}$, $\Lambda_t^{ESS,bu}$ show the operation costs and power exchange of ESS units, Λ_t^{IBDR} shows the IBDR programs implementing cost, Λ_t^{Load} shows the loads operation costs and $\Lambda_t^{EV,se}$, $\Lambda_t^{EV,bu}$ show the operation costs and power exchange of EV unit. In Eq. (12-6), the EV participation coefficient δ_p^{IBDR} represents the effective share of EV participation in IBDR-based L2G services. This coefficient is evaluated based on key EV behavioral and operational factors, including the state of charge, initial charging condition, charging-time requirement, and mobility-related usage pattern. Therefore, EV participation in the proposed model is not treated as unconditional full compliance, but as a state-dependent and constrained contribution. This interpretation is consistent with recent studies showing that EV flexibility and schedulability are strongly shaped by user charging decisions, SoC limits, and arrival-departure characteristics [27,28]. $P_t^{grid,se}$, $P_t^{grid,bu}$ show the power exchanged between the MG and the main grid, P_t^{WT} , P_t^{PV} show the extracted power from the PV and WT units, $P_t^{ESS,se}$, $P_t^{ESS,bu}$ show the power exchanged by the ESS unit, P_t^{Up} , P_t^{Down} the shifted-up or shifted-down powers in the IBDR programs, P_t^{Load} , ΔP_t^{Load} show the operation power of the load and its changes, and $P_t^{EV,se}$, $P_t^{EV,bu}$ express the power exchanged by the EVs. Also δ^{IBDR} indicates the participation coefficient of each effective component on the first level objective function based on the IBDR program requirements. The symbol ($\pm$) indicates that the exchange of power is two-way. N_{EV} indicates the EVs number participating in IBDR programs. Eqs. (12-8) to (12-12) show the constraints, upper and lower boundaries of different components. According to (11), the second level of the objective function includes risk assessment and creating a safe cyberspace.

$$f_{Risk,t}^{MG} = \sum_{t=1}^{T=24} \left(\frac{(C_t^{Risk} \cdot RIF_t^{Risk}) \cdot \sum_{s \neq 0}^S C^A \cdot \Delta PDF_t^{Data}}{C^C} \right) \quad (13)$$

$$\begin{cases} 0 < C^A, C^C < 1 & C^C \leq RIF_t^{Risk} \leq C^A \\ \Delta PDF_t^{Data} \leq \Delta PDF_t^{Data} \leq \Delta PDF_t^{Data} \end{cases} \quad (14)$$

Eq. (13) is the execution cost of the safe region, where C_t^{Risk} represents the cost of the safe region and risk assessment, RIF_t^{Risk} represents the risk impact factor, C^A , C^C are the accuracy and confidence coefficient of the data, respectively, and ΔPDF_t^{Data} expresses the probability distribution function deviation coefficient of the input data. It has been mentioned in Eq. (11) that the third level of the objective function involves the implementation of L2G services. Equations are formulated to minimize active and reactive power deviation subject to constraints of voltage stability and frequency.

$$f_{L2G,t}^{MG} = \left[\begin{aligned} & \left(\sum_{t=1}^{T=24} \int_0^T (\zeta_f \cdot \delta_f) \cdot (P_t^{ref} - P_t^{Op}) dt \right) + \left(\sum_{t=1}^{T=24} \int_0^T (\zeta_v \cdot \delta_v) \cdot (Q_t^{ref} - Q_t^{Op}) dt \right) \\ & + \sum_{t=1}^{T=24} \theta^{ADF} \cdot f_t^{MG,n} + \sum_{t=1}^{T=24} \theta^{ADV} \cdot V_t^{MG,n} \\ & \text{Voltage and frequency stability} \end{aligned} \right] \quad (15)$$

$$0 \leq \theta^{ADF} \leq 0.01 f_t^{MG,n} \quad (16)$$

$$0 \leq \theta_t^{ADF} \leq 0.05 V_t^{MG,n} \quad (17)$$

$$\begin{aligned} \Delta f &= (f_{ref} - f_{op}) \begin{cases} (\delta_f) = 1; & \text{if } |\Delta f| > 0.4 \text{ Hz} \\ (\delta_f) = 0; & \text{if } |\Delta f| < 0.4 \text{ Hz} \end{cases} \\ \Delta V &= (V_{ref} - V_{op}) \begin{cases} (\delta_v) = 1; & \text{if } |\Delta V| > 0.05 \text{ pu} \\ (\delta_v) = 0; & \text{if } |\Delta V| < 0.05 \text{ pu} \end{cases} \end{aligned} \quad (18)$$

According to (15), L2G services are formulated to reduce active and reactive power deviation as well as the participation of MGs in regulating the voltage and frequency, where ζ_f , ζ_v indicate the frequency and voltage deviation, respectively, δ_f , δ_v describe the frequency and voltage deviation, P_t^{ref} , P_t^{Op} are the reference and operation active power, Q_t^{ref} , Q_t^{Op} are the reference and operation reactive power, $f_t^{MG,n}$ indicates the nominal frequency of MG, $V_t^{MG,n}$ represents the nominal voltage of MG and θ^{ADF} , θ^{ADV} indicate the permissible coefficients of voltage and frequency deviation, respectively. Eq. (18) shows the voltage and frequency deviation, where Δf , ΔV describe frequency and voltage fluctuations, f_{ref} , f_{op} describe the reference and operation frequency, and V_{ref} , V_{op} show the reference and operation voltage.

2.4. IBDR program configuration

IBDR programs have enabled subscribers to participate in L2G services in a competitive energy market. These programs benefit both the DSO and subscribers. The study is focused on IBDR programs and price elasticity theory, which exhibit either mutual-elasticity or self-elasticity in response to price changes [23,25].

$$P_{n,t}^{IBDR} = \delta_p^{IBDR} \cdot P_{n,t}^0 \cdot \left[\begin{aligned} & \left[\Delta P_{n,t}^{IBDR} \cdot (1 + E_{ti}) \cdot \left(\frac{\Lambda_{n,t}^{spot} - \Lambda_{n,t}^0}{\Lambda_{n,t}^0} \right) \right] \\ & + \sum_{i \neq j}^{24} \left[\Delta P_{n,t}^{IBDR} \cdot (1 + E_{ij}) \cdot \left(\frac{\Lambda_{n,t}^{spot} - \Lambda_{n,t}^0}{\Lambda_{n,t}^0} \right) \right] \end{aligned} \right] \quad (19)$$

Eq. (19) shows the IBDR program model based on the self and mutual-elasticity, where $P_{n,t}^{IBDR}$ has expressed IBDR-based power changes, $P_{n,t}^0$ shows the initial power of the loads, $\Delta P_{n,t}^{IBDR}$ shows the power changes in IBDR program, E_{ti} shows the self-elasticity, E_{ij} shows the mutual-elasticity, $\Lambda_{n,t}^{spot}$ is energy market spot price, and $\Lambda_{n,t}^0$ is the energy market base price. For more details, the equations and proof of the IBDR program are detailed in Appendix 2.

2.5. Target points and hybrid cyber-attack detection algorithm

The cybersecurity layer considered in this study is defined on the communication-driven interaction among microgrid components, where hackers target the exchanged supervisory and operational signals rather than only the physical devices themselves. In this framework, hybrid cyber-attacks (FDI/DoS) are modeled as the combined effect of false data injection and communication-disruption behavior over the

MG communication network. More specifically, FDI alters the numerical content of exchanged measurements or control-related signals, while DoS-type interference degrades the continuity, timeliness, or effective availability of communication, thereby increasing the mismatch between reference and operational information across different layers of the microgrid. Accordingly, the attack surface in this work is the set of data paths connecting market, control, and operational entities, and the main objective of the attacker is to distort the information environment

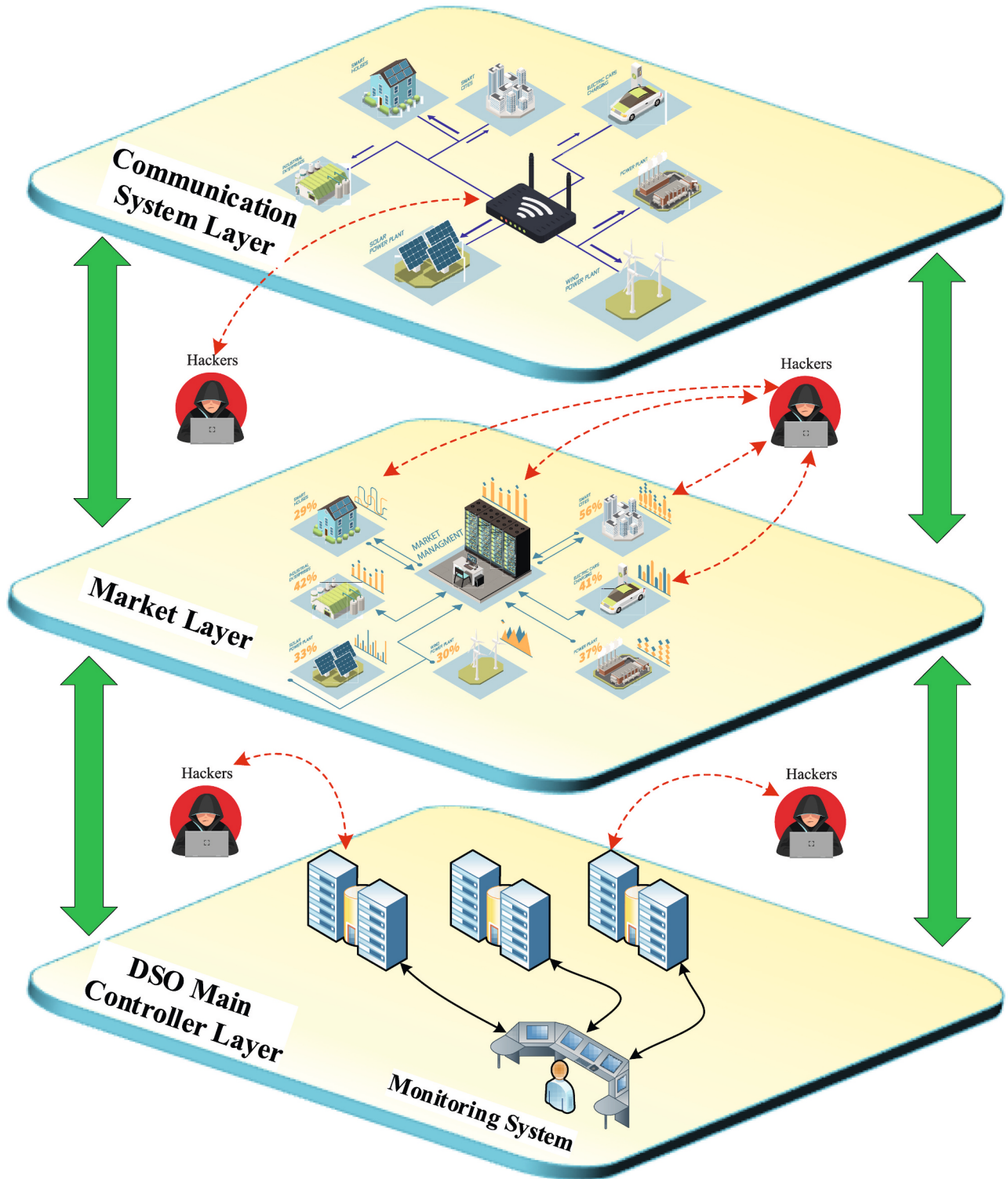


Fig. 1. Layered structure of MG and target points of hybrid cyber-attacks.

in a way that can mislead decision-making, destabilize L2G-support actions, and degrade the integrity of trained control policies.

Based on this representation, the proposed detection mechanism is designed as a multi-stage dynamic data evaluation process. First, the exchanged signals are monitored at the communication level and compared with their reference counterparts in order to quantify the inconsistency introduced by malicious penetration. Then, the attack effect is transferred to the measurement-state layer through a weighted least-squares residual formulation, so that abnormal deviations are evaluated not only as local signal errors but also as state-relevant inconsistencies within the nonlinear MG structure. Finally, a safe-region concept is constructed by combining Gaussian and Bernoulli probabilistic modeling with Euclidean-distance evaluation in spherical space. This allows the method to capture both continuous abnormal drift and discrete attack events, while also providing an interpretable geometric boundary between normal and malicious data. Therefore, the detection framework in this section is intended to move from signal corruption, to inconsistency quantification, to residual-based validation, and finally to safe-region classification of cyber-attacked data.

According to Fig. 1, hackers can influence different MG layers by using communication systems and uploading false information based on error vectors. In the following, the modeling of cyber-attack penetration points and an attack detection approach based on a dynamic data evaluation framework are presented [8,22].

$$M^{H-Attack} = \begin{bmatrix} S_{1,1}^{H-Attack} & \dots & S_{1,j}^{H-Attack} \\ \vdots & \ddots & \vdots \\ S_{i,1}^{H-Attack} & \dots & S_{i,j}^{H-Attack} \end{bmatrix} \quad (20)$$

$$S_{ij}^{MG} = S_{ij}^{Com} + S_{ij}^{H-Attack} \quad \underbrace{S_{ij}^{H-Attack} = s_{ij}^{H-Attack} f^{H-Attack}}_{H-Attack \text{ Vector}} \quad (21)$$

$$I^{H-Attack} = \chi^{H-Attack} \cdot \left[\sum_{i=1, j \neq 1} \left(S_{Ref,ij}^{output} - S_{OP,ij}^{output} \right) \right] \quad (22)$$

$$\begin{cases} \text{if } I^{H-Attack} < I_{min}^{H-Attack} \Rightarrow S_{ij}^{H-Attack} = 0 \\ \text{if } I^{H-Attack} > I_{min}^{H-Attack} \Rightarrow S_{ij}^{H-Attack} \neq 0 \end{cases} \quad (23)$$

Eq. (20) shows the hybrid cyber-attack matrix, where $M^{H-Attack}$ is the hybrid cyber-attack matrix and $S_{ij}^{H-Attack}$ is the arrays of the hybrid cyber-attack matrix. Eq. (21) shows the signals exchanged in the MG structure, where S_{ij}^{MG} expresses the signal exchanged in the MG structure between nodes i and j , S_{ij}^{Com} expresses the actual signal of the communication system, and $S_{ij}^{H-Attack}$ represents the vector of cyber-attacks loaded on actual signal of the communication system. The vector of cyber-attacks also has two components, where $s_{ij}^{H-Attack}$ shows the scaling factor of cyber-attacks penetration and $f^{H-Attack}$ describes the fundamental penetration factor. Eq. (22) shows the inconsistent term in different layers of MG, where $\chi^{H-Attack}$ describes the inconsistent coefficient and $S_{Ref,ij}^{output}$, $S_{OP,ij}^{output}$ describe the deviation coefficient of the reference and operation output signal in the communication system, respectively. Eq. (23) also shows that if the inconsistent coefficient is lower than the predetermined value, then the injected data is not considered as cyber-attacks. If the inconsistent coefficient is higher than the values predefined, then the cyberattacks vector is then loaded on to the communication system's master vector. For discovering the dynamic nature of cyberattacks, weighted least squares error methodology was utilized here [22]. According to (24), if an x -dimensional measurement matrix equivalent to M_x is defined based on the nonlinear components MG (Z), then the measurement matrix M_x includes the matrix of y -dimensional state variables equivalent to M_y and the error matrix (M_e).

$$M_x = \begin{bmatrix} (ZM_y + M_e)_{1,1} & (ZM_y + M_e)_{1,j} \\ (ZM_y + M_e)_{i,1} & (ZM_y + M_e)_{i,j} \end{bmatrix}_{i \times j} \quad (24)$$

$$\hat{M}_y = \underset{M_y}{\operatorname{argmin}} J(M_y) = (M_x - ZM_y)^T \cdot W(M_x - ZM_y) \quad (25)$$

According to (25), the nature of the dynamic variables affecting the state vector caused by cyber-attacks is described as a residual function based on the weighted least square error method, where $\hat{M}_y$ is the state vector matrix affected by cyber-attacks and $J(M_y)$ is the residual function of the weighted least square method. In this equation, W is the weight matrix resulting from the variance of the probability distribution functions related to the measurement data and includes $\operatorname{diag}\{\delta_1^{-2}, \delta_2^{-2}, \dots, \delta_{M_x}^{-2}\}$.

Theorem 1. *The dynamic nature of cyber-attacks is detected by analyzing measurement data against predetermined values. If the left side of the inequality exceeds the inconsistent coefficient, cyber-attacks are present in the measurement matrix. The range of predetermined values can be determined using the Euclidean norm, Gaussian probability distribution function, and Bernoulli distribution, allowing interpretation of the dynamic nature of data in spherical space [29,30].*

$$J(\hat{M}_y) = (M_x - Z\hat{M}_y)^T \cdot W(M_x - Z\hat{M}_y) > I_{min}^{Attack} \quad (26)$$

Proof 1. Gaussian function of distribution, based on Euclidean distance and Bernoulli distribution, is used to check dynamic variables and a safe area can be designated in a sphere space to check malicious data [29,30]. Given that dynamic information is inside the safe area, it's treated as unaffected by cyberattacks. But if information is beyond the safe area, the information will be considered cyberattacks. If marginal data (on the spherical surface) lead to a standard deviation of the Gaussian distribution function (above 5%), then the threshold $I_{min}^{H-Attack}$ is achieved and such data are regarded as error variables.

Statement. : The dynamic nature of cyber-attacks in microgrid systems can be effectively determined by comparing measurement data with values established a priori in terms of statistical distributions and geometric parameters. More specifically, cyber-attacks are detected if the measurement data strays out of a specified "safe region," which is calculated based on Gaussian distribution, Bernoulli distribution, and Euclidean distance. The secure region is such that it keeps valid operational data within a spherical region, and data beyond this region are recognized as malicious. The method relies on examining dynamic variables according to a combination of probability distribution functions and distance metrics.

Assumptions. : The assumptions provided below are to model mathematical equations.

1. Measurement data in the microgrid system is subject to cyber-attacks and is prone to generating errors in data.
2. The system has a Gaussian distribution function that describes the anticipated continuous behavior of data when there is no attack.
3. There exists a Bernoulli distribution function that describes the discrete attack-based anomalies or outliers.
4. Detection of attacks is based on deviations from a reference signal, using the Euclidean distance to decide whether data is within or outside the safe region.

In the present study, the attacker is modeled as an adversarial entity acting through the communication layer of the microgrid and targeting the exchanged signals among market, control, and operational components. The studied hacker is assumed to have the ability to inject malicious dynamic data into communicated measurements and supervisory

signals, thereby altering their content and degrading the consistency between reference and operational information. Accordingly, the hybrid attack vector considered here represents the combined effect of FDI-type signal manipulation and communication-disruption behavior that can distort the observation space used by the control framework. The attacker is therefore assumed to influence decision making indirectly through corrupted information rather than through direct physical destruction of devices. Under this assumption, the attack is detected through its measurable impact on exchanged signals, residual-based state inconsistency, and deviation from the admissible safe region defined by the probabilistic and geometric detection model.

The evidence relies on three fundamental elements: the Gaussian distribution function, Bernoulli distribution, and Euclidean distance. These are utilized to determine the safe area where valid data ought to be, and any deviation outside this area signifies a cyber-attack. The Gaussian distribution determines the likelihood of a data point being valid based on how close it is to the center of the distribution. The standard deviation impacts to quantify the spread of the data, values a specified number of standard deviations from the mean to be normal. This is stated as:

$$G(x) = 0.05 \cdot \alpha_i \cdot \left[\frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2} \right] \quad (27)$$

where $G(x)$ is the probability density function (PDF), μ is the mean, and σ is the standard deviation of the data. Data points x that are distant from the mean (i.e., beyond a safe limit, typically set at $\mu \pm 3\sigma$) are marked as suspected attacks. The Bernoulli distribution is applied to model discrete events which may be either a success or a failure, which in the event of binary attack detection is attack/no attack. It is used to identify whether certain attack patterns occur in the data, that is, to detect outliers or sudden injection anomaly patterns:

$$\begin{cases} \text{Prob}\{\alpha_i(k) = 1\} = \bar{\alpha}_i \Rightarrow \text{Attacks} \\ \text{Prob}\{\alpha_i(k) = 0\} = 1 - \bar{\alpha}_i \Rightarrow \text{No attacks} \end{cases} \quad (28)$$

where $\bar{\alpha}_i$ is the probability of a specific data point belonging to an attack event. Spikes or variations in the data that are unusual are described by the Bernoulli distribution to isolate the events that depict the cyber-attacks. The Euclidean distance metric is used to calculate the distance of a data point from the center of the safe region anticipated. If the data point lies outside this region, it would be suspicious. For data points in a high-dimensional space (e.g., voltage, frequency), the Euclidean distance from the true measured value (z) to the predicted safe measured value (w) is given by:

$$\begin{cases} d(z, w) = (1 - \bar{\alpha}_i) \sqrt{(z_1 - w_1)^2 + (z_2 - w_2)^2 + \dots + (z_n - w_n)^2} = (1 - \bar{\alpha}_i) \cdot \|z, w\|_2 \\ \text{Threshold} \Rightarrow J(\hat{M}_y) = (M_x - Z\hat{M}_y)^T \cdot W(M_x - Z\hat{M}_y) \text{ or } (1 - \bar{\alpha}_i) \cdot \|z, w\|_2 > I_{\min}^{H-\text{Attack}} \end{cases} \quad (29)$$

where z is the observed data vector and w is the reference (or expected) data vector. $\|\cdot\|_2$ is the Euclidean norm. The data is revealed to be outside the safe area if the distance is larger than a specific threshold, which indicates a possible cyber-attack. The detection criterion is whether the residual value, as determined by the weighted least squares error method, is larger than a certain threshold. The threshold is derived from the standard deviation of the Gaussian distribution and the sensitivity of the Bernoulli distribution. If the residual is larger than $I_{\min}^{H-\text{Attack}}$, a cyber-attack is identified. The safe region of genuine data exists in a sphere with the point w at the center, and any transition out of this

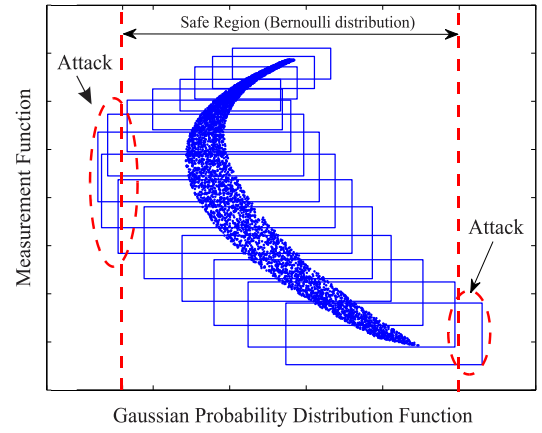


Fig. 2. Discrete dynamic data detection based on Gaussian distribution.

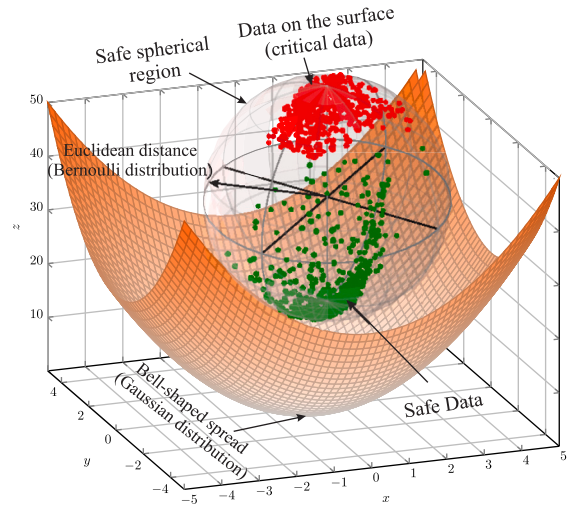


Fig. 3. Safe region evaluation and behavior of continuous dynamic data.

sphere is classified as malicious. The safe region is bounded by the radius $d(z, w)$ via the Gaussian distribution and Euclidean distance.

Figs. 2 and 3 are crucial for understanding the process of detection of dynamic data in the event of cyber-attacks in the microgrid system. Fig. 2 shows how the detection of dynamic data is done using the

combination of the Gaussian distribution and the Bernoulli distribution. In this plot, the process of detection is shown, marking how normal data are within the “safe region” and any suspicious or malicious data points are outside, indicated as potential attacks. The Gaussian distribution is used to model the behavior of normal data under the assumption that the system's operation is governed by a general probabilistic distribution, and the Bernoulli distribution is used to highlight discrete anomalies in this data set. By combining these, the system can detect data anomalies that would indicate cyber-attack activity.

Fig. 3 shows the assessment of the safe region in spherical space, based on the Euclidean distance. It employs a spherical space model

where the assessment of Euclidean distance between measured data and reference data is carried out. Data points within this spherical space are considered “safe,” i.e., they are within the tolerated ranges of system behavior. When the data points go out of the sphere, they are signaled as anomalies with the presence of potential cyberattacks. This spatial approach helps visualise how such cyberattacks will cause deviations in normal operation by effectively highlighting such security threats imposed by such an attack. Figs. 2 and 3 provide robust frameworks for dynamic data analysis of cyber-attack detection. With statistical models, like Gaussian and Bernoulli distributions, combined with distance measures such as Euclidean distance, the system ensures that there is an operational space secured, where the malicious interventions causing deviations can be quickly detected and combated effectively.

Algorithm 1 summarizes the complete workflow of the proposed hybrid cyber-attack detection framework. It begins with the initialization of the microgrid communication structure and the construction of the hybrid attack matrix, then monitors the exchanged signals among communication nodes to identify deviations from the nominal information flow. After computing the inconsistency coefficient, the method builds the measurement and state matrices and evaluates cyber-induced abnormalities through a weighted least-squares residual function. Finally, by combining Gaussian and Bernoulli probabilistic interpretation with Euclidean-distance-based safe-region assessment, the algorithm classifies the measured data as either safe or malicious. Therefore, Algorithm 1 provides an integrated procedural description of how hybrid FDI/DoS-type cyber-attacks are detected, validated, and isolated within the proposed BMADRL-based cybersecurity framework.

Algorithm 1. Pseudocode for hybrid cyberattack detection.

Input: Microgrid communication structure $MG_{structure}$, exchanged signal matrix S , hybrid cyber-attack matrix M

Output: CyberAttack_Flags, SafeData_List, MaliciousData_List

- 1 Initialize $MG_{structure}$
- 2 Initialize signal matrix S
- 3 Initialize hybrid cyber-attack matrix M
- Step 1: Model hybrid cyber-attack matrix (Eq. 20)
- 4 Construct the attack-vector matrix model
- Step 2: Monitor exchanged signals in communication system (Eq. 21)
- 5 **foreach** node i in $MG_{structure}$ **do**
- 6 **foreach** neighbor j in $Neighbors(i)$ **do**
- 7 $S_{exchanged}[i][j] \leftarrow S_{actual}[i][j] + S_{cyber-attacks_loaded}[i][j]$
- Step 3: Compute inconsistency coefficient (Eq. 22)
- 8 **foreach** signal path (i, j) **do**
- 9 $I[i][j] \leftarrow \text{inconsistency_coefficient}[\sum |S_{ref}[i][j] - S_{op}[i][j]|]$
- Step 4: Cyber-attack identification logic (Eq. 23)
- 10 **foreach** (i, j) **do**
- 11 **if** $I[i][j] > \text{threshold_value}$ **then**
- 12 Mark $M[i][j]$ as CyberAttack
- 13 **else**
- 14 $M[i][j] \leftarrow 0$ No attack detected
- Step 5: Build measurement matrix (Eq. 24)
- 15 $M_x \leftarrow \text{MeasurementMatrix}$
- 16 $M_y \leftarrow \text{StateVector}$
- 17 $M_e \leftarrow \text{ErrorMatrix}$
- Step 6: Compute residual function using WLS (Eq. 25)
- 18 Compute the dynamic variables affecting the state vector caused by cyber-attacks
- Step 7: Dynamic detection via statistical evaluation (Eq. 26)
- 19 **if** $\text{ResidualFunction} > \text{Inconsistency_Coefficient}$ **then**
- 20 Flag the measurement as attacked
- 21 **else**
- 22 Mark the measurement as normal
- Step 8: Define safe region in spherical space (Eqs. 27–29)
- 23 **foreach** data point z in MeasurementData **do**
- 24 Evaluate GaussianDistribution
- 25 Evaluate BernoulliDistribution
- 26 Compute $d_{Euclidean}$
- Step 9: Safe-region classification
- 27 **if** $d_{Euclidean} \leq \text{safe_radius}$ **then**
- 28 Label the data as SafeData
- 29 **else**
- 30 Label the data as MaliciousData
- 31 **Return** CyberAttack_Flags, SafeData_List, MaliciousData_List

3. Microgrid robust defense mechanism based on BMADRL

3.1. MADRL framework

The BMADRL method consists of two basic frameworks, which include MADRL and BDRL. MADRL is described by using Markov decision process (MDP) equations as well as the Bellman entropy theory [31]. To configure the MDP equations, first, the state space equations are formulated and then by deriving from the state space equations, the MDP equations can be extracted. Hence, the state space equations are:

$$\begin{aligned} \dot{x}(t) &= f(x_t, a_t, d_t) \\ y(t) &= h(x_t, a_t, d_t) \end{aligned} \quad (30)$$

where x_t shows the system state vector (e.g., voltage, frequency, control signal states), a_t describe action/control input from MADRL, d_t is disturbance due to cyber-attacks (FDI/DoS), $y(t)$ is output vector (measurable system variables), $f(\cdot)$ is system dynamics, possibly nonlinear and data-driven, $h(\cdot)$ is output function. Let $x_t \in \mathbb{R}^n$ be the state vector representing MG components (e.g., bus voltages, frequency deviations, SoC of ESS, RER output power, etc.). Let $a_t \in \mathbb{R}^m$ be the action vector from the BMADRL agent (e.g., ESS control signals, load adjustments, power setpoints for EVs and RERs). Let $d_t \in \mathbb{R}^n$ be the disturbance vector from cyber-attacks, modeled as injected error signals. The system's nonlinear dynamic evolution is approximated as:

$$\dot{x}(t) = f(x_t, a_t + g(d_t)) \quad (31)$$

where $f(\cdot)$ is the BMADRL-controlled system model approximated by neural networks using MDP dynamics, and $g(d_t)$ represents the cyber-attack injection, where FDI changes signal values and DoS affects signal delays or losses. The following structure can be used for observation model with cyber-attack detection integration.

$$y(t) = h(x_t, d_t) = x_t + d_t \quad (32)$$

Let $y(t) \in \mathbb{R}^p$ be the observed/measured outputs, where in FDI, d_t is an additive vector corrupting measurements, in DoS, d_t introduces delay/missing data patterns. To derive the MDP equations from the state space equations (Eq. 31), we have: an MDP is formally defined as a tuple (S, A, P, R) where, S is the set of states (e.g., voltage, frequency, RER/ESS states), A is the set of actions (e.g., control setpoints for loads, ESS, EVs), $P(s'|s, a)$ is transition probability to state s' given s, a , R is reward function. To implement the discrete-time state transition equation to MDP equations, consider the following equation. Discretizing Eq. (31) for time steps $t = 0, 1, 2, \dots, t$ we get:

$$x_{t+1} = f(x_t, a_t) + d_t \quad (33)$$

where $x_t \in S$ is state set at time t , $a_t \in A$ is action set taken at time t , and d_t is cyber disturbance (modeled as noise or probabilistic injection). This equation defines the transition kernel $P(x_{t+1}|x_t, a_t)$, where the stochastic component is caused by d_t (e.g., FDI or DoS-induced randomness). The control action is derived from the optimal policy π^* using Q-learning with Bayesian belief updating:

$$a_t = \pi^*(x_t, b_t) \quad (34)$$

where π^* is control policy and b_t is the belief update from Bayesian inference (refer to Eq. 39). To make the learning mechanism more explicit, the instantaneous reward is formulated as a weighted negative function of the main operational and cyber-secure objectives of the microgrid. In this way, the BMADRL agent is encouraged to reduce the operating cost and cyber-related risk, preserve the effectiveness of L2G services, and avoid unsafe conditions caused by malicious dynamic

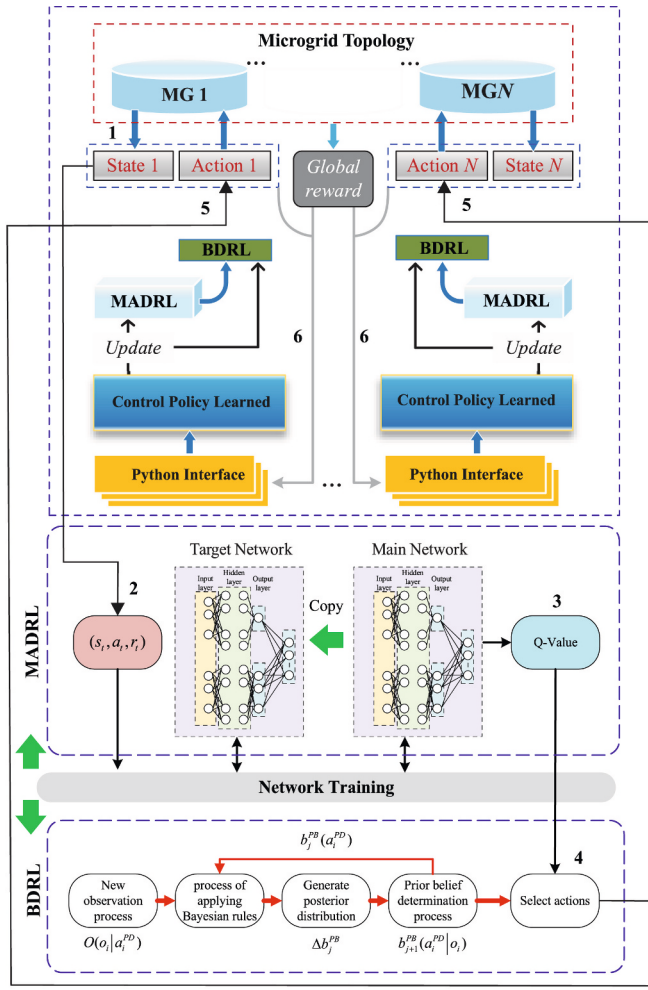


Fig. 4. Performance of the BMADRL approach.

deviations. The reward function is expressed as

$$r_t = -[\lambda_1 J_t^{\text{op}} + \lambda_2 J_t^{\text{risk}} + \lambda_3 J_t^{\text{L2G}} + \lambda_4 \max(0, I_t^{\text{H-Attack}} - I_{\min})] \quad (34a)$$

where r_t denotes the instantaneous reward at time t , J_t^{op} is the operating-cost component, J_t^{risk} is the risk-related component, and J_t^{L2G} is the L2G-service performance component, all derived from the multi-level objective function defined in Eq. (11). The term $I_t^{\text{H-Attack}}$ represents the cyber inconsistency coefficient associated with abnormal deviations in the communication and measurement space, as described in Eqs. (22) and (23), while $I_{\min}$ is the minimum admissible threshold used to distinguish secure behavior from malicious deviation. Accordingly, the last term penalizes cyber-induced inconsistencies that exceed the safe operating boundary. The coefficients λ_1 , λ_2 , λ_3 , and λ_4 are nonnegative weighting factors satisfying $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1$, and they determine the relative importance of economic performance, cyber resilience, and secure L2G service delivery. Based on the above reward definition, the learning structure of the MADRL approach can be expressed through the Q-learning update rule in the DQN framework. In this formulation, the agent updates its action-value estimate according to the immediate reward obtained from secure economic operation and the expected long-term return of the next state [30].

$$Q_{\text{new}}(s_t|a_t) = \left(Q_{\text{old}}(s_t|a_t) + \theta \left(r_t + \tau \max_{a_t} Q(s_{t+1}|a_t) - Q_{\text{old}}(s_t|a_t) \right) \right) \quad (35)$$

where θ expresses the learning rate, τ represents the discount factor, $Q_{\text{new}}(s_t|a_t)$ and $Q_{\text{old}}(s_t|a_t)$ are the new and old Q-value, r_t is the compo-

nent of the reward set in Eq. 34a and $Q(s_{t+1}|a_{t+1})$ describes and the Q-value of the new state in the new episode. To describe the learning network based on the target network, the loss function based on (36) can be used.

$$F_{\text{MADRL}}^{\text{Loss}} = E(r_t + \tau \max_{a_t} Q((s_{t+1}|a_t), W') - Q((s_t|a_t), W)) \quad (36)$$

where $F_{\text{MADRL}}^{\text{Loss}}$ expresses the loss function of the MADRL approach, E is the loss function coefficient, and W, W' express the weight function of the main and target networks. The $(r_t + \tau \max_{a_{t+1}} Q((s_{t+1}|a_{t+1}), W'))$ describes that the target Q-value is generated by the target network, and the $Q((s_t|a_t), W)$ also describes that the current Q-value is predicted by the main network. However, in Deep Q-Network (DQN), the max operator selects overestimated values, which leads to overoptimistic estimation [32]. For this purpose, Double DQN (DDQN) architecture is proposed, where the selection of actions and their evaluation are separated [33]. DDQN loss function is defined as follows.

$$F_{\text{MADRL}}^{\text{Loss-DDQN}} = E(r_t + \tau Q(s_{t+1}, \arg\max_{a_t} (s_{t+1}|a_t), W') - Q((s_t|a_t), W)) \quad (37)$$

The target network evaluates the actions and the main network selects actions based on $\arg\max_{a_t} (s_{t+1}|a_t), W)$. By separating the selection and evaluation of actions, DDQN can provide more accurate Q values than DQN.

3.2. BDRL framework

The BDRL framework uses observations to update beliefs based on Bayesian rules [34,35]. It supports and verifies trained policies in the MADRL approach. If an agent is attacked, the agents cannot track the state and action set. Attacked agents can apply relevant actions based on observations. The BDRL unit serves as a backup based on past experiences.

$$b_j^{PB}(a_i^{PD}|o_i) = \frac{O(o_i|a_i^{PD}) \cdot b_j^{PB}(a_i^{PD})}{\sum_{a_i \in A} O(o_i|a_i^{PD}) \cdot b_j^{PB}(a_i^{PD})} \quad (38)$$

where a_i^{PD} expresses the posterior distribution of the action in the agent under cyber-attack, o_i expresses the observations, $b_j^{PB}(a_i^{PD}|o_i)$ expresses the expected probability for choosing the action a_i^{PD} based on the observations o_i , $b_j^{PB}(a_i^{PD}|o_i)$ is a conditional probability based on observations that cannot be used as a new belief for the next update [34,35]. To update the new belief for future episodes, the following equation can be used.

$$D_j^{\text{New}}(b_j^{PB}(a_i^{PD})) = \frac{c_{ij-1} + b_j^{PB}(a_i^{PD}|o_j)}{\sum_{i=1}^{|A_i|} c_{ij-1} + b_j^{PB}(a_i^{PD}|o_j)} = \frac{c_{ij-1} + \Delta b_j^{PB}}{\sum_{i=1}^{|A_i|} c_{ij-1} + \Delta b_j^{PB}} \quad (39)$$

where $D_j^{\text{New}}(b_j^{PB}(a_i^{PD}))$ is the new probability distribution for action a_i^{PD} , Δb_j^{PB} is the new update for action selection, c_{ij-1} is also an estimated coefficient for action selection.

In the BDRL design, action selection is directly coupled with the observation space. This relationship is illustrated in Fig. 4. As shown in the figure, the BMADRL framework is organized into two stages. The control policies represented by the Q-values of the MADRL model are evaluated against the outputs of the BDRL model in order to assess the extent to which policy degradation is reduced during offline training episodes and during control policy testing. The operational flowchart involves several steps. The initial state sets are sent to the MADRL unit, followed by evaluating the initial state and action sets in network training. The initial Q-value is produced, and configured in the BDRL process, and actions are selected based on the BDRL structure. Control

policy signals are generated with a new set of actions, and actions are valued by the reward pool. This process continues until the offline training episodes are completed. During online operation, executive policies are compared with the actions component confirmed by the BDRL approach, and the BDRL unit prevents the possible destruction of these policies. Fig. 4 is numbered for a better understanding. Initially and in step 1, the system state is sampled and a state set is generated. Then in step 2, the state set, action set and reward set based on MDP equations are configured. In the structure of the MADRL model, there are two networks, namely the main network and the target network, where the target network is responsible for evaluating the actions and the main network is responsible for selecting the actions. In step 3, after selecting the action in the main network, the selected action is compared with the action models in the history of the BDRL approach, which is

BDRL unit then validates whether the candidate action remains consistent with trusted observation-supported behavior before it is applied to the microgrid. If the action is validated, it is executed in the system transition; otherwise, it is corrected through the BDRL backup mechanism. The reward evaluation, loss computation, gradient-based update of the main network, and belief-distribution update then complete the decision cycle. In this way, Algorithm 2 makes explicit that belief updating is embedded inside the real-time control loop itself and that the interaction between MADRL and BDRL is realized through sequential action generation, Bayesian validation, execution filtering, and belief propagation across episodes and agents.

Algorithm 2. Pseudocode for MG Robust Defense Mechanism Based on BMADRL.

```

1: Initialize state set  $S_t$   $\triangleright$  Includes voltage, frequency, SoC of ESS, RER outputs, etc. (Eq. 30)
2: Sample initial environment state vector  $x_t \in S_t$ 
3: Define action set  $A_t$   $\triangleright$  e.g., load adjustments, ESS controls (Eqs. 30–33)
4: Define reward set  $R_t$   $\triangleright$  Based on economic cost, risk, and L2G service performance
5: Initialize  $Q_{\text{main}}$  and  $Q_{\text{target}}$  networks (MADRL-DDQN)
6: for each training episode do
7:   for each agent in the microgrid do
8:     Select action  $a_t$  using  $Q_{\text{main}}$ 
9:      $b_t \leftarrow \text{Bayesian\_Update}(\text{Prior}_{b_t}, \text{Observation\_History})$ 
10:     $\text{Posterior}_{a_t} \leftarrow \text{Bayesian\_Inference}(a_t, b_t)$ 
11:    if BDRL\_Validate( $\text{Posterior}_{a_t}, \text{Observation\_History}$ ) then
12:      Apply  $a_t$  to MG control policy
13:    else
14:       $a_t \leftarrow \text{Adjust\_Action}(\text{BDRL\_Backup\_Model})$ 
15:    end if
16:     $S_{t+1} \leftarrow \text{Transition}(S_t, a_t, d_t)$ 
17:    reward  $\leftarrow \text{Evaluate\_Reward}(S_t, a_t)$ 
18:    Loss  $\leftarrow \text{Compute\_Loss}(Q_{\text{target}}, Q_{\text{main}})$ 
19:    Update  $Q_{\text{main}}$  weights using gradient descent
20:    Update belief distribution over  $A_t$  using latest observations
21:  end for
22: end for
23: Share experience trajectories among agents
24: Propagate belief updates across the agent network
25: Save optimized and validated policy  $\pi^*$ 
26: Deploy policy for online operation (real-time control)

```

shown in step 4. If the selected action is free from illegal deviations, then this action will be applicable to the microgrid topology. As shown in step 5, the action that matches the trusted models in the BDRL approach is sent to the microgrid topology and used. In step 6, the reward set, which is responsible for valuing the actions according to the operational accuracy of each action, improves the process of updating the behavior pattern of the actions in the MADRL model to configure the best actions selection between the MADRL and BDRL approaches. Bayesian formulation provides principled estimation of uncertainty within historic data and thus reduces reliance on potentially biased and unrepresentative samples. Moreover, the approach provides generalizability preserved by learned policies through simulation of diverse adversarial scenarios during training and posterior regularization techniques. Coordination among inter-agents reduces decision-making variance too by repeatedly calibrating policies based on common belief and experience. Together, these traits create a robust learning paradigm that can maintain high performance despite the conditions differing from the training conditions. For more support, Algorithm 2 illustrates the structure of Section 3.

Algorithm 2 provides a clearer procedural interpretation of the integration between MADRL and BDRL within the BMADRL framework. It shows that the control process begins with state initialization and MDP-based definition of the state, action, and reward sets, after which the MADRL layer generates a candidate action through the main Q-network. This action is not executed directly; instead, it is passed to the Bayesian layer, where prior belief and observation history are used to construct an updated belief state and a posterior action assessment. The

4. Experimental configuration and results

As illustrated in Fig. 5, the modeling and experimental platform is established on the IEEE 69-bus test system, operating at a rated voltage of 400 V and a nominal frequency of 50 Hz [36]. The study environment covers several integrated components, including microgrid representation on the standard IEEE 69-bus network using RT-LAB 2021.3 and HYPERSIM 2021.3, a real-time simulation platform supported by the code generator and OP5707XG hardware, and a cyber-communication layer developed in EXata CPS v1.1 in accordance with IEC 61850 protocol specifications [36,37]. Whereas, the experimental setup environment includes equipment such as a main controller, oscilloscope, communication infrastructure in different layers (NI9205), and also data acquisition system. For the experimental setup, the total load demand is set at 10 MW, while the capacities of the renewable energy resources and energy storage system units are each fixed at 1 MW. The participation increment for the incentive-based demand response program is defined as 100 kW. In addition, the EV battery model is specified using actual KIA battery data, where each unit is characterized by a rated voltage of 697 V, a rated power of 253 kW, an energy capacity of 77.4 kWh, and an approximate charging current of 110 A [22]. To support data handling and control execution in the laboratory platform, the TMS320F28335 processor is employed. The converter and filtering parameters are also defined according to the experimental configuration. In particular, the PWM switching frequency is selected as 15 kHz, and the output filter is modeled with an LC structure of 1.9 mH and 35 μ F, while maintaining total harmonic distortion below 5% [22]. A summary

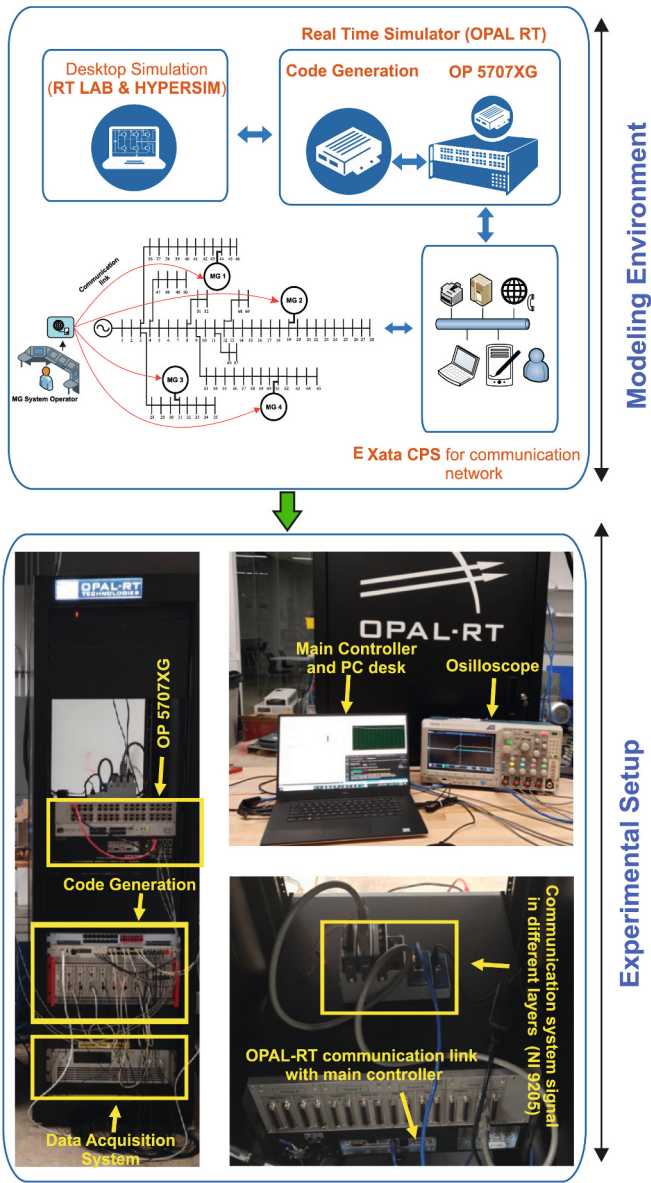


Fig. 5. MG cyber space configuration.

Table 2
BMADRL processing components.

Component	Value	Component	Value
Learning rate	$1e^{-3}$	Discount factor	0.959
Sampling	5000	Entropy weight	0.0212
Optimizer type	Adam	Rate of Update	0.0004
Validation process	21,000	Testing process	21,000
Training process	80,000	Overall process	122,000
Offline training time	1.36×10^4 (s)	Online operation time	1(s)

of the processed data associated with the proposed method, including both the offline training stage and the online operational stage, is reported in Table 2.

4.1. Experimental results

The experimental results and the BMADRL approach effectiveness against cyber-attacks for minimizing their impacts are presented in two case studies. The first stage evaluated cyber-attack effects on L2G service

Table 3
Operation of MGs with/without hybrid cyber attacks.

Hours	Scenario I (\$)			Scenario II (\$)		
	$f_{OP,t}^{MG}$	$f_{Risk,t}^{MG}$	$f_{L2G,t}^{MG}$	$f_{OP,t}^{MG}$	$f_{Risk,t}^{MG}$	$f_{L2G,t}^{MG}$
1.	977.0	337.6	467.7	862.3	305.5	407.5
2.	1008.1	347.1	484.3	889.7	314.1	422.0
3.	1076.0	387.8	507.1	949.7	350.8	441.8
4.	1110.6	396.0	522.5	980.2	358.4	455.3
5.	1128.5	412.7	557.1	996.0	373.5	485.4
6.	1138.0	428.3	578.6	1004.4	387.5	504.1
7.	1146.4	412.7	579.7	1011.9	373.5	505.2
8.	1147.7	411.6	587.0	1012.8	372.4	511.5
9.	1160.7	429.5	597.6	1024.4	388.6	520.8
10.	1339.6	719.4	598.9	1182.4	650.9	521.9
11.	1326.6	730.1	600.1	1170.8	660.6	522.8
12.	1318.3	727.7	609.6	1163.5	658.5	531.2
13.	1355.2	743.1	612.0	1196.1	672.6	533.2
14.	1342.1	731.3	603.6	1184.5	661.7	526.0
15.	1135.7	437.8	576.1	1002.3	396.1	502.1
16.	1136.9	437.8	567.8	1003.4	396.1	494.8
17.	1395.8	847.0	632.3	1231.9	766.5	551.0
18.	1410.1	849.4	639.4	1244.5	768.6	557.1
19.	1431.6	859.0	646.5	1263.4	777.2	563.4
20.	1449.4	862.5	676.4	1279.3	780.5	589.4
21.	1431.6	859.0	665.6	1263.4	777.2	580.1
22.	1375.6	726.5	598.9	1213.9	657.4	521.9
23.	1336.2	720.5	600.0	1179.2	652.0	522.8
24.	1319.4	695.5	594.0	1164.4	629.4	517.6
Daily	29,997	14,509	14,102	26,474	13,129	12,288
Total	58,608 (\$)			51,891 (\$)		
Hacker profit	6717 (\$)			0		

policies to increase hackers' profit in the energy market. While, the second stage assessed the cyber-attack effects on critical indexes of MGs connected to the main grid, such as voltage and frequency are explained in below sections.

4.1.1. First operation stage results

Table 3 reports the operating results of the first microgrid under two different conditions. Scenario I represents the case in which cyberattacks are neither detected nor mitigated, while Scenario II corresponds to the case in which attack-induced errors in the L2G-enabled energy market are identified and eliminated. The table also presents the aggregated 24-h operating cost of the microgrid. The total daily cost reaches 58,608 \$ in Scenario I, whereas it declines to 51,891 \$ in Scenario II. These results confirm that the presence of unmitigated cyberattacks imposes a clear economic burden on system operation, leading to an 11.46% increase in total operating cost relative to the protected case. In addition, the attacker gains 6717 \$ by manipulating market price and demand signals. A more detailed examination of Table 3 together with Eq. (11) further shows that the first cost component of the objective function ($f_{OP,t}^{MG}$) is significantly improved once the proposed defense mechanism is activated. Specifically, under Scenario II this part of the objective function decreases by 11.75% compared with Scenario I. This reduction indicates that the proposed method not only suppresses malicious market

Table 4
Comparison of the proposed method with DRL and RL under cyberattack scenarios.

Scenario	Model	$f_{OP,t}^{MG}$	$f_{Risk,t}^{MG}$	$f_{L2G,t}^{MG}$	Total cost (\$)	Hacker profit (\$)
Scenario 1	Proposed Method	29,997	14,509	14,102	58,608	6717
	DRL	31,647	15,525	15,019	62,191	8140
	RL	32,997	16,250	15,653	64,900	9565
Scenario 2	Proposed Method	26,474	13,129	12,288	51,891	0
	DRL	28,062	14,114	13,271	55,447	1268
	RL	29,651	14,836	14,008	58,495	2194

interference but also improves the economic performance of the microgrid by restoring more reliable operational decisions under L2G policies.

Also, the second term of the objective function ($f_{Risk,t}^{MG}$) has decreased by 9.5% when compared to Scenario I, and the third term of the objective function ($f_{L2G,t}^{MG}$) has decreased by 9.2%. Therefore, the results show that the proposed approach is important in reducing the operating expenses of MGs because it does not destroy control policies and also eliminates cyberattacks without detection. Nevertheless, the intrusion of hackers in the layers of the energy market gives rise to an unrealistic scenario to maximize the profit of hackers. The intrusion of hackers results in a reduction of the involvement of subscribers in L2G services, which in turn results in a tremendous escalation of operation and maintenance costs in power systems. On the other hand, as the extent of subscriber participation decreases, DSO must structure power management schemes in accordance with the upstream grid, thereby driving the price upward in energy market exchanges.

Table 4 shows a clear performance advantage of the proposed method over the DRL and RL baselines in both operating conditions. Under Scenario 1, where no cyberattack mitigation is applied, the proposed method yields the lowest operational objective cost, objective function cost, and L2G service objective cost, leading to the minimum total daily cost of 58,608 \$. By comparison, the total cost rises to 62,191 \$ for DRL and 64,900 \$ for RL, indicating that the proposed framework provides more efficient scheduling and stronger economic coordination even before the activation of any defense mechanism. A similar trend is observed in the hacker profit indicator, where the proposed method limits the attacker's gain to 6717 \$, whereas DRL and RL allow higher

profits of 8140 \$ and 9565 \$, respectively. These results suggest that the proposed model is inherently more robust against adversarial disturbances and less vulnerable to exploitative manipulation.

A more important distinction appears in Scenario 2, where cyber-attack mitigation is activated. In this case, all methods improve their economic performance, but the proposed method still achieves the best result, reducing the total cost to 51,891 \$, compared with 55,447 \$ for DRL and 58,495 \$ for RL. More importantly, the hacker profit is reduced to zero only for the proposed method, while residual attacker benefits remain for DRL and RL at 1268 \$ and 2194 \$, respectively. This difference should not be interpreted as an unfair comparison, since the attack scenario, operating conditions, and evaluation criteria are kept identical for all methods. However, the mitigation capability is not assumed to be equally effective across the compared models; rather, it is treated as part of each method's own decision and defense performance. Therefore, the results indicate that RL and DRL can only partially reduce the impact of cyberattacks, whereas the proposed method is capable of fully neutralizing the attacker's economic benefit while maintaining the lowest operating cost.

4.1.2. Second operation stage results

Here, the efficiency of BMADRL against cyberattacks on the control elements of the MG is explained. The approach has been hailed to be rapid, accurate, and to catch and eradicate bad dynamic data and thus realize an essential step to prevent destruction of control policies. Fig. 6 (a, b) demonstrates the performance of the BMADRL controller against cyber-attacks affecting the output power of RERs and the absorbed power of ESS units. The cyber-attacks occur at $t = 1$ s, and the controller

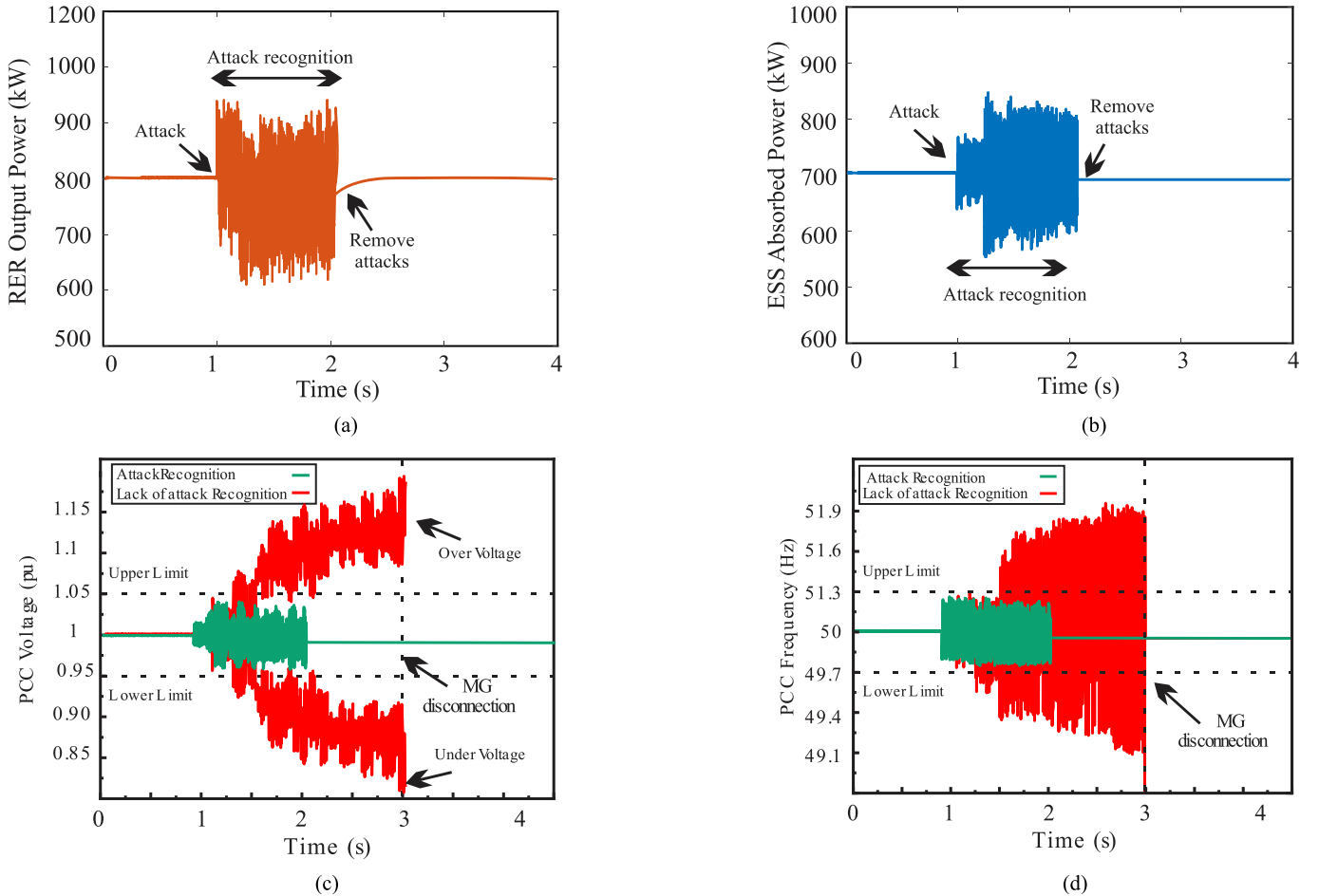


Fig. 6. Effects of cyber-attacks on different MG components (a) RER output power oscillation (b) ESS absorbed power oscillation (c) PCC voltage oscillation (d) PCC frequency oscillation.

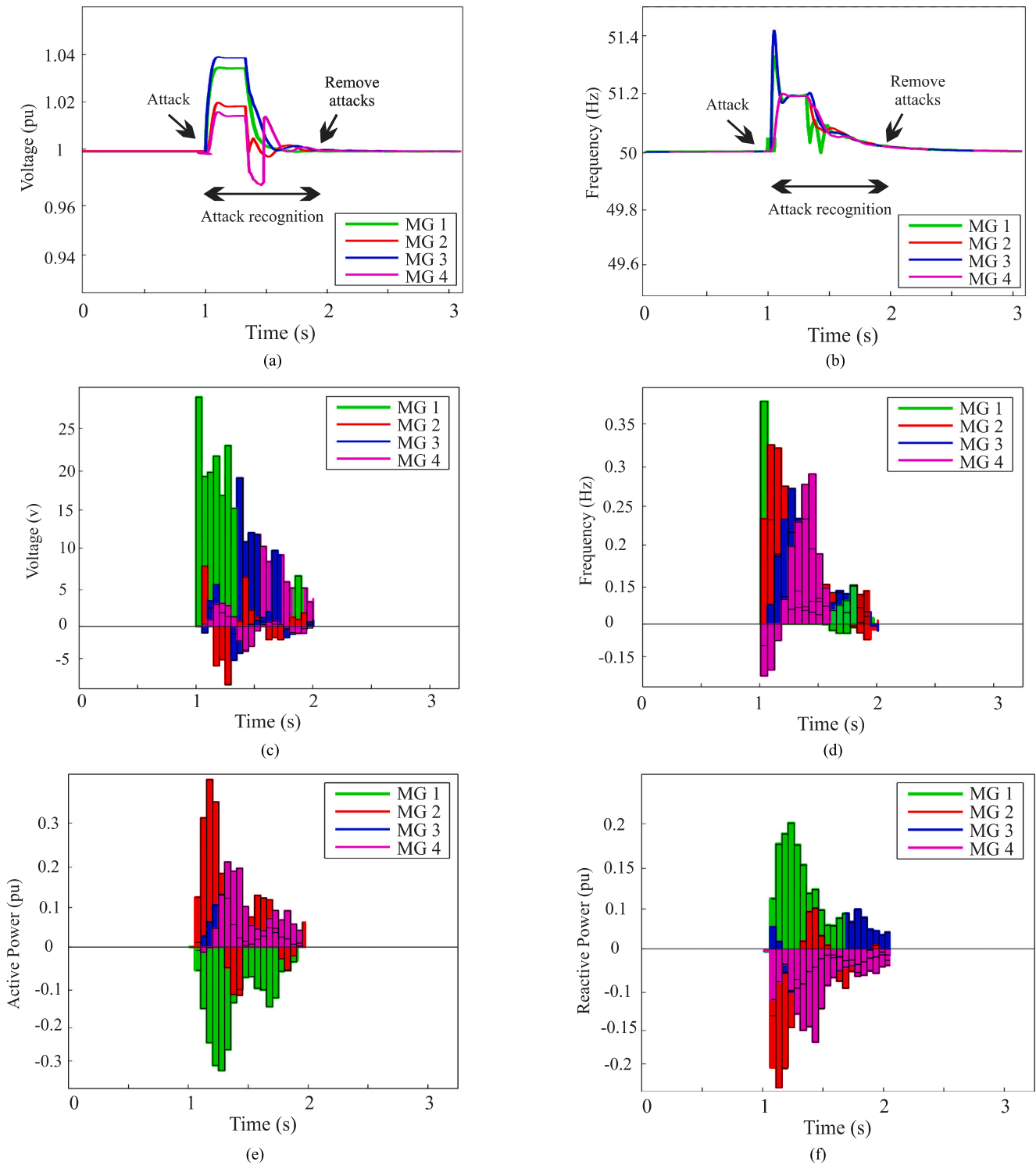


Fig. 7. (a,b) Voltage and frequency control of different MGs, (c,d) changes of voltage and frequency components by the proposed controller, (e,f) changes of active and reactive power components by the proposed controller.

successfully stabilizes the MG after detecting and removing the attacks after 1-s. According to Fig. 6 (c, d), the BMADRL approach effectively detects, removes, and maintains MG stabilization, while also showing the lack of detection effects of cyber-attacks. Failure to promptly detect cyber-attacks leads to severe voltage and frequency oscillations, resulting in MG disconnection from the point of common coupling (PCC). The proposed approach with a 1 s operating interval is crucial for maintaining MG transient stability.

In the following, the effectiveness of the BMADRL method has been confirmed with respect to the damage due to simultaneously launched cyber-attacks on various MGs. According to Fig. 7 (a, b), $t = 1$ s is the instant when cyberattack initiates. The mentioned technique meets the transient stability requirements of the MG at $t = 2$ s by removing malicious dynamic variables using detection. Since the voltage and frequency oscillations are regulated within a short time interval, the transient stability constraints as a function of the admissible values of

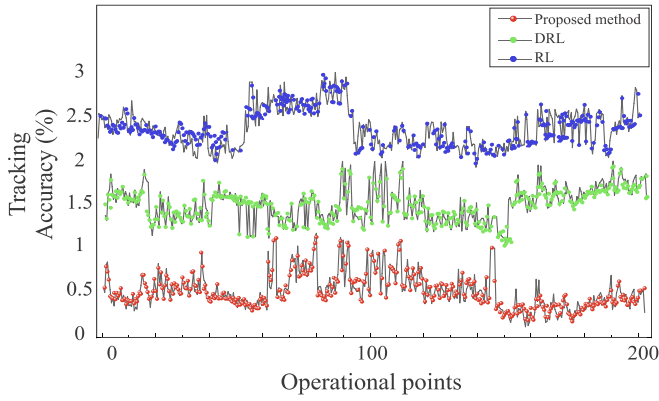


Fig. 8. Tracking cyber attacks.

the voltage and frequency oscillations are properly monitored. Therefore, for in-depth analysis, according to Fig. 7 (c, d), the continuous voltage and frequency component oscillations as a function of the operating performance of the proposed controller are shown. The developed controller here can control the voltage and frequency by adjusting active and reactive power components in MGs. Therefore, according to Fig. 7 (e, f), variations in net active and reactive powers of each MG unit are shown to control PCC's voltage and frequency. According to Fig. 7, the used methodology successfully detects, removes, and guarantees transient stability against hybrid cyber-attacks. It prevents erasing control policies with high efficiency, accuracy, and extremely high speed of operation. It satisfies transition stability constraints and improves MG security against cyberattacks.

Fig. 8 clearly demonstrates the capability of the proposed solution to monitor hybrid cyber-attacks over 200 varied operating points in comparison with the DRL and RL methods. The results show that the BMADRL algorithm maintains an average tracking tolerance of less than

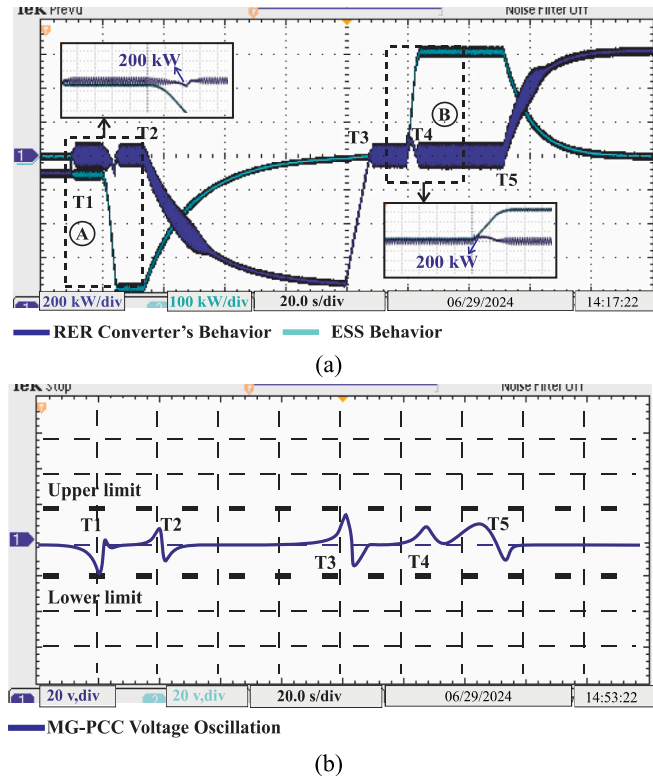


Fig. 9. (a) Coordinated operation of RER and ESS units, (b) satisfaction of PCC voltage stability constraints.

0.5%, whereas the corresponding average tolerance rises to 1.5% for DRL and 2.5% for RL. This result provides direct quantitative support for the superior tracking precision of the proposed framework in following malicious dynamic variations under different operating conditions. The stronger performance of BMADRL can be attributed to the coordinated structure of the MADRL and BDRL modules, where the learning process is supported by probabilistic policy verification and online validation of suspicious deviations. As a result, the proposed framework preserves more reliable tracking behavior and reduces the impact of cyber-induced distortions more effectively than the benchmark DRL and RL approaches across diverse microgrid operating scenarios. These quantified results also provide a clearer numerical basis for interpreting the tracking accuracy of the proposed method under attack-aware operation.

L2G service framework based on the IBDR program is necessary to ensure the operation stability and agility of MGs, particularly for power quality regulation and economic performance. By enabling dynamic load response and optimized energy trading, the IBDR program ensures that distributed energy resources like RERs and ESSs play an active role in system regulation. Fig. 9 (a, b) illustrates coordination between RER and ESS units, portraying their complementary features in voltage support at the PCC. Coordination is particularly important in responding to the innate intermittency of RERs such as photovoltaic and wind units whose generation is fluctuating depending on environmental conditions. The ability of the ESS units to respond instantly to these variations by injecting or absorbing power is of utmost significance in maintaining voltage stability. The coordination of the same has been well presented in the tested T₂ and T₅ operating points. At T₂, a decrease in RER generation triggers the ESS unit to release stored power onto the grid to compensate for the loss to keep the voltage levels at the PCC within acceptable margins. At T₅, a rise in RER generation, however, enables the ESS units to reduce their injection and instead utilize the surplus power to avoid over-voltage and systemwide inefficiencies. Such behavior is an example of a smart, responsive control system that compliments the goals of L2G services in today's MGs. Moreover, the scheduling coordination of IBDR-based L2G allows for predictive as well as real-time power balancing, reducing dependence on the upstream grid and system stress. Also, coordinated operation enhances the economic effectiveness of MGs by providing optimal utilization of energy storage, extending ESS lifetimes, and reducing curtailment of the RER. As is depicted in Fig. 9 (b), owing to this interaction between RER and ESS units there occurs systematic satisfaction of the PCC voltage limitations under unbalanced operation conditions, thereby promoting the microgrid system robustness and reliability. The above result reaffirms that the developed control strategy not only supports voltage compensation but also strengthens greater objectives towards energy efficiency, sustainability, and market participation in accordance with the IBDR program.

4.2. Sensitivity assessment

Sensitivity analysis is used here to examine the robustness of the control framework under a wide range of operating conditions. In this study, the proposed method is evaluated against RL and DRL over 200 operating points. These points reflect diverse microgrid states, including variations in unit demand profiles, grid-connected and islanded transitions, fluctuations in PV and battery output, disturbances in the energy market, and uncertainties within the communication layer. As shown in Fig. 10, the proposed approach maintains the most stable response across these operating points. Its average sensitivity to operating changes is around 2%, whereas the corresponding values for RL and DRL remain above 5%, indicating that the proposed framework is less affected by operating disturbances and changing system conditions.

The same trend appears in the offline training stage, as illustrated in Fig. 11. The proposed method yields a noticeably lower sensitivity error rate than both RL and DRL during policy training. More specifically, its

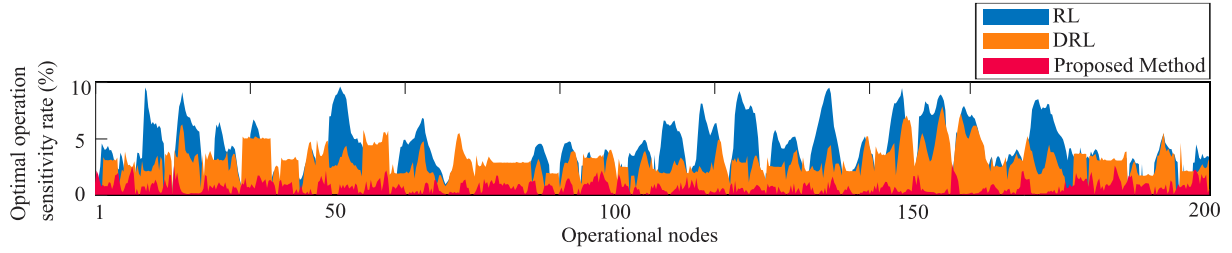


Fig. 10. Optimal operating sensitivity rate.

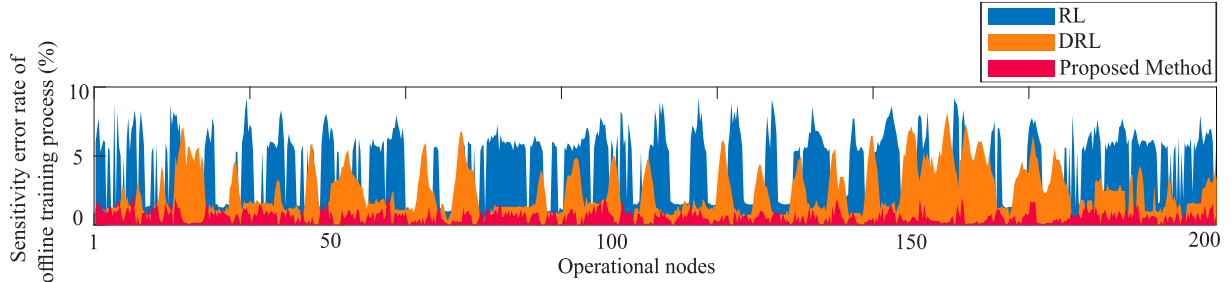


Fig. 11. Sensitivity error rate in offline training process.

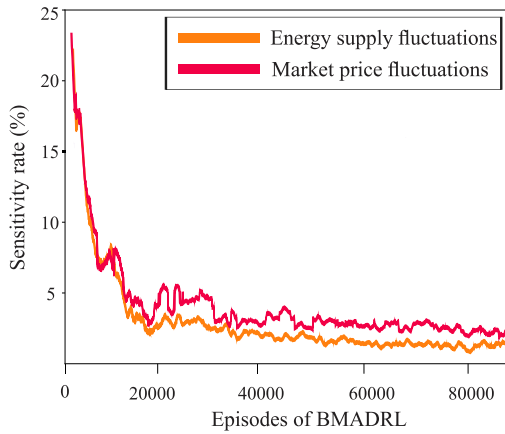


Fig. 12. Evolution of uncertainty sensitivity during offline policy learning.

average training sensitivity remains below 2%, while the error level for RL and DRL generally falls within the 5% to 10% range. This result suggests that the proposed framework is more effective in preserving the consistency of learned control policies during offline training and is less prone to policy degradation. From this perspective, the method extends

conventional machine learning-based control by improving resistance to cyberattack-related disturbances and better protecting the integrity of the trained decision policies.

Fig. 12 indicates that the proposed BMADRL framework progressively suppresses the influence of key uncertainty sources during training, particularly those associated with energy market volatility and variations in power supply. This behavior reflects a gradual improvement in model calibration, where repeated interaction with uncertain operating conditions helps stabilize the learned control response. Across 80,000 training episodes, the observation-based learning mechanism not only updates the control policy but also preserves its consistency as training evolves. As a result, the final policy becomes better suited to operate under diverse and uncertain system conditions. By the end of the training process, the sensitivity of the proposed model to both supply-side fluctuations and market disturbances drops to below 2%, which provides strong evidence of the method's effectiveness and robustness.

Fig. 13 further confirms the computational advantage of the proposed method over the DRL and RL benchmarks. The results show that the computational error rate of the proposed framework is 2%, whereas the corresponding values increase to 6% for DRL and 8% for RL. This result is consistent with the lower operating sensitivity and offline-training sensitivity reported in Figs. 10 and 11, and indicates that the proposed BMADRL structure preserves more stable numerical behavior

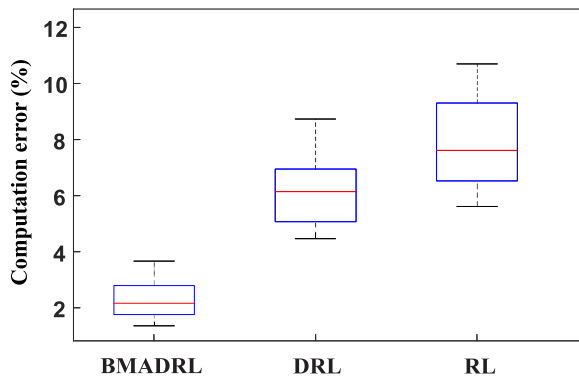


Fig. 13. Relative computational error performance of the compared methods.

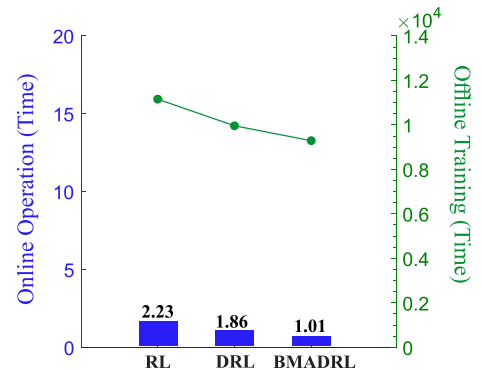


Fig. 14. Comparative behavior of the proposed framework in the training and deployment stages.

Table 5
Quantitative performance indicators.

Metric	Proposed method	DRL	RL	Unit	Evidence source
Offline training time	0.9×10^4	0.9×10^4	0.9×10^4	s	Fig. 14
Online operation time (including detection stage)	1.01	1.86	2.23	s	Fig. 14
Tracking tolerance over 200 operating points	0.5	1.5	2.5	%	Fig. 8
Computational error rate	2.0	6.0	8.0	%	Fig. 13
Operating sensitivity rate	2.0	5.0–10.0	5.0–10.0	%	Fig. 10
Offline-training sensitivity error	2.0	5.0–10.0	5.0–10.0	%	Fig. 11

under cyberattack-aware operation. The lower computational error also suggests that the learned control policies remain more coherent and less affected by cyber-induced distortions than those obtained by the DRL and RL methods.

Fig. 14 complements this result by comparing the offline training and online operation processes of the three methods. The proposed method requires 0.9×10^4 s for offline training, compared with 1.0×10^4 s for DRL and 1.15×10^4 s for RL. In the online stage, the proposed framework operates in 1.01 s, while the corresponding online operation times are 1.86 s for DRL and 2.23 s for RL. These results provide direct quantitative evidence that the proposed method imposes a lower computational burden in both the learning and execution stages.

The shorter online operation time is particularly important in the present application, because the online process includes the detection stage and directly affects how quickly the control framework can react to cyber-induced deviations. A faster execution cycle enables the proposed method to activate corrective and defensive actions more rapidly, thereby reducing the risk of propagation of malicious dynamic variables through the microgrid. Therefore, the combined evidence from Figs. 13 and 14 shows that the proposed method is not only computationally more efficient than DRL and RL, but also better suited for time-sensitive cyber-secure microgrid operation.

Table 5 provides a consolidated quantitative summary of the main performance indicators of the proposed method and the benchmark DRL and RL approaches. As shown in the table, the proposed framework achieves the shortest offline training time and online operation time, while also maintaining the lowest tracking tolerance, computational error rate, operating sensitivity, and offline-training sensitivity error among the compared methods. These results show that the proposed BMADRL-based structure is not only faster in both learning and online execution, but also more accurate and numerically stable under cyberattack-aware operation. Therefore, Table 5 offers a direct

quantitative basis for interpreting the claims related to response speed, computational efficiency, and tracking performance in the experimental section.

Fig. 15 shows the power curve along with the operational graph of the DR program that modifies the consumption pattern of subscribers such as ESS and EV units. According to Fig. 15(a), the power price curve has three characteristics, namely low price, middle price and high price at peak consumption. Accordingly, the DR program is configured according to the power price curve and frames the components of changes in the consumption pattern of subscribers. Therefore, according to Fig. 15(a) and the blue curve, the approach of the DR program can be seen. In this approach, an attempt has been made to flatten the load profile so that not only the power system (reducing peak power consumption) benefits from this pattern, but also subscribers gain more profit by changing their consumption pattern. Therefore, according to this figure, subscribers are encouraged to match their power according to the DR pattern. For example, according to Fig. 15(b), ESS and EV units purchase power during low or middle prices and consume it during peak periods to avoid paying more for power purchase. This approach reduces peak consumption and flattens the load profile. The behavior pattern of ESS and EV units tracks the changes in the DR graph, and hence subscribers increase their profit function by purchasing power during off-peak periods and consuming it during peak periods.

5. Conclusion

This study was motivated by the increasing dependence of modern microgrids on digital communication infrastructures, market-oriented coordination, and machine-learning-based control, all of which have expanded system flexibility but also increased vulnerability to hybrid cyber-attacks. In particular, false data injection and denial-of-service-type disturbances can degrade communication reliability, destabilize energy management decisions, and compromise the integrity of trained control policies during both offline learning and online operation. Since L2G service provision increasingly depends on the coordinated participation of ESS units, EVs, responsive loads, and renewable resources, the protection of these control policies becomes essential not only for cybersecurity purposes but also for preserving normal market participation and operational stability.

To address this challenge, this paper proposed a robust cyber-secure defense framework based on Bayesian multi-agent deep reinforcement learning. The proposed BMADRL structure combines the adaptive decision-making capability of MADRL with the probabilistic validation and policy-support function of BDRL in order to reduce the degradation of trained control policies under malicious dynamic data conditions. In parallel, a safe-region-based hybrid cyber-attack detection mechanism was formulated by integrating Gaussian and Bernoulli probabilistic modeling, Euclidean-distance-based evaluation, and residual analysis through the weighted least-squares method. This joint design enabled the framework to detect abnormal cyber-induced deviations, verify

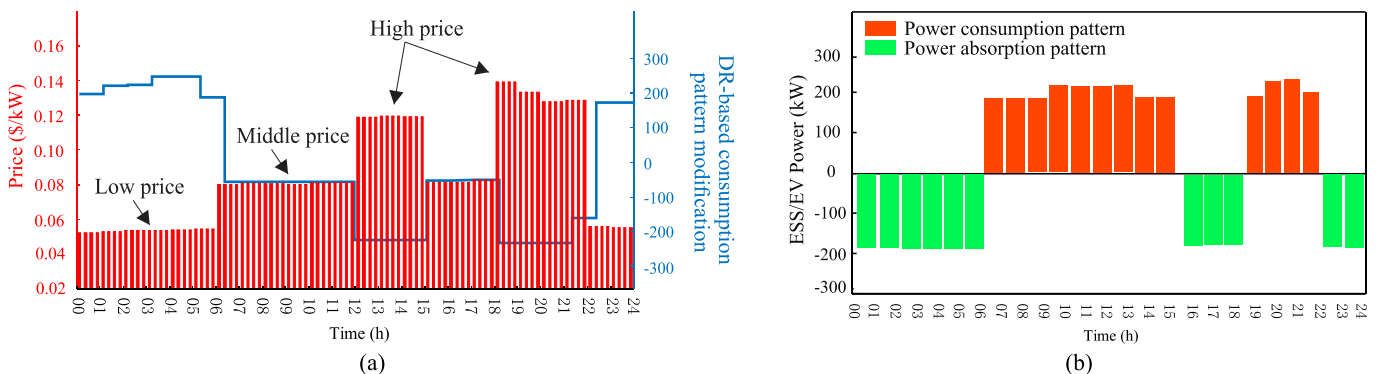


Fig. 15. Power price curve and modification of ESSs/EVs consumption pattern.

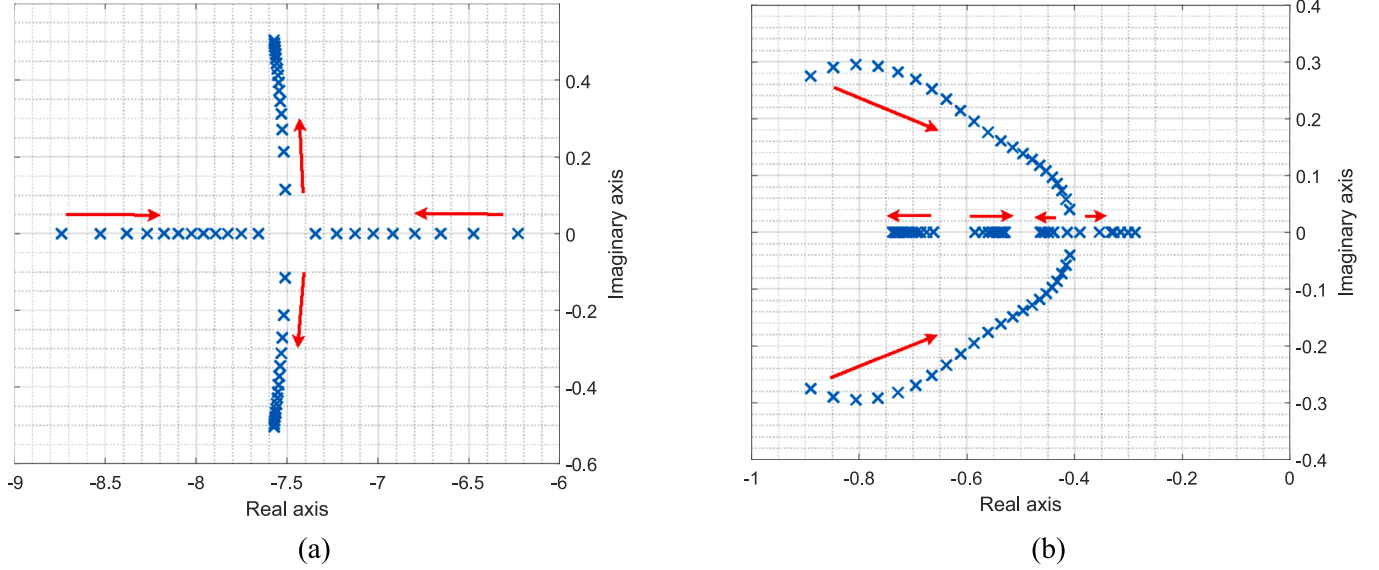


Fig. 16. Root locus diagram for variation of Net power.

action consistency, and preserve secure L2G-oriented operation in the presence of hybrid attack conditions.

The results confirmed that the proposed method provides meaningful economic and operational benefits. In the protected operating scenario, the BMADRL framework reduced the total operating cost by 11.46% and eliminated the attacker's economic gain, while also preserving secure energy market interaction. In addition, the proposed method maintained a tracking tolerance below 0.5% across 200 operating points, which indicates strong cyber-attack monitoring capability under diverse operating conditions. The time-domain results further showed that the framework could mitigate attack effects within approximately 1 s and prevent severe voltage and frequency oscillations, thereby supporting transient stability and secure microgrid operation.

The quantitative comparison with the DRL and RL benchmark methods further strengthened the validity of the proposed framework. The proposed method achieved the lowest computational error rate, the shortest offline training time, and the shortest online operation time among the compared methods, while also exhibiting lower operating sensitivity and lower offline-training sensitivity error. These results indicate that the superiority of the BMADRL framework is not limited to attack detection alone, but also extends to numerical reliability, learning stability, and time-sensitive cyber-secure execution. Therefore, the proposed structure offers a more balanced combination of economic efficiency, response speed, and policy-protection capability than the benchmark RL- and DRL-based approaches.

Despite these contributions, one limitation of the present study is that the framework was validated on a specific microgrid configuration and under the studied hybrid cyber-attack setting, so the reported

performance should be interpreted within this experimental scope. In particular, although the proposed method demonstrated strong economic, computational, and cyber-secure control performance, its scalability and behavior under larger, more heterogeneous, and more communication-intensive microgrid environments still require further investigation. Accordingly, a meaningful future research direction is to extend the BMADRL framework to larger-scale cyber-physical energy systems with richer network topologies and broader uncertainty sources in order to further evaluate its generalizability and practical deployment capability.

CRedit authorship contribution statement

Reza Sepehrzad: Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Milad Moafi:** Writing – review & editing, Visualization, Validation, Formal analysis. **Nima Khosravi:** Writing – review & editing, Validation, Investigation, Formal analysis, Data curation. **Ahmed al Durra:** Validation, Supervision, Project administration, Funding acquisition, Conceptualization. **Mahdieh S. Sadabadi:** Visualization, Validation, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1

Stability assessment

For further support, the stability assessment is first configured according to the Lyapunov stability equations and then visually illustrated by the root locus method [38,39]. The following are the different steps of this stability assessment.

Step 1: Let the dynamics of the microgrid + control policy be described by [13]:

$$\dot{x}(t) = f(x_t, a_t) + d_t \quad (40)$$

where x_t shows the system state (voltage, frequency, control policy state, etc.), a_t describe the control action from MADRL, d_t describe disturbance due to cyber-attack (error vector injected via FDI or DoS) and $f(\cdot)$ shows nonlinear function approximated by MADRL policies.

Step 2: Formulate a Lyapunov function in terms of state deviation from nominal (safe) state x^* :

$$V(x) = \frac{1}{2}(x - x^*)^T P (x - x^*) \quad (41)$$

where $P = P^T > 0$ is a positive-definite matrix and x^* is the desired equilibrium or setpoint (can include learned optimal policy Q-values).

Step 3: Through the acquisition of the Lyapunov candidate, the stability condition can be established and verified.

$$\dot{V}(x) = (x - x^*)^T P \dot{x} = (x - x^*)^T P [f(x, a) + d_t] \quad (42)$$

To ensure stability, we want $\dot{V}(x) < 0$. Let's analyze this with two terms which are normal conditions and no risks from cyberattacks and hybrid cyberattack conditions [5,15]. In the normal conditions if the main controller (MADRL) is well trained, then the stability can be guaranteed by referring to the condition presented in the previous equation [40]. From this we have:

$$\dot{V}(x) = (x - x^*)^T P f(x, a) < 0 \quad (43)$$

While the operation conditions are considered taking into account the conditions of hybrid cyberattacks, then the physical body of the derivative of the candidate Lyapunov equation is configured as follows:

$$\dot{V}(x) = (x - x^*)^T P f(x, a) + (x - x^*)^T P d_t \quad (44)$$

Step 4: To guarantee robustness, the BMADRL method introduces the Bayesian policy validation (limits large deviations) and Safe-region enforcement (detects and removes $d(t)$). If $d(t)$ is **bounded** and the Gaussian-Bernoulli safe region keeps $\|d_t\| < \delta$, then:

$$\dot{V}(x) \leq -\alpha \|x - x^*\|^2 + \beta \|x - x^*\| \cdot \|d_t\| \quad (45)$$

where $\alpha, \beta > 0$ show design constants and δ is threshold radius of the safe region (hyper-sphere boundary).

Step 5: For sufficiently small $\|d_t\|$, Lyapunov stability is preserved. This leads to **Input-to-State Stability (ISS)**:

$$\dot{V}(x) \leq -\alpha \|x - x^*\|^2 + \beta \|d_t\|^2 \quad (46)$$

When the BMADRL policy guarantees the system remains in a well-defined bounded region, and cyber-attack perturbations are rapidly detected and effectively neutralized, the microgrid resilience is significantly enhanced. The system is then maintained with Lyapunov stability, regardless of the dominance of interfering, bounded cyberattacks. This not only guarantees operation reliability but also validates the robustness of control policies against dynamically evolving cyber vulnerabilities (Fig. 16).

Appendix 2

A2.1 IBDR model considering changes in subscriber power due to price elasticity

The price elasticity of demand is the sensitivity of subscriber demand to price changes [41]. Based on this definition, we can write:

$$E_t = \begin{cases} E_{i,i} = \frac{\Lambda_i^0}{P_i^0} \cdot \frac{\partial P_i}{\partial \Lambda_i} \\ E_{i,j} = \frac{\Lambda_j^0}{P_i^0} \cdot \frac{\partial P_i}{\partial \Lambda_j} \end{cases} \quad (47)$$

where E_t indicates the price elasticity, $E_{i,i}$ indicates the self-elasticity, $E_{i,j}$ describes the cross-elasticity, Λ_i^0 indicates the base price of the energy carrier in the i_{th} group, P_i^0 is the base power of the i_{th} group, $\frac{\partial P_i}{\partial \Lambda_i}$ is the rate of change of the i_{th} group power based on price changes, Λ_j^0 is the price of the energy carrier in the j_{th} group, and $\frac{\partial P_i}{\partial \Lambda_j}$ is the rate of change of the power in the i_{th} group based on price changes of the j_{th} group.

Confronted with volatile electricity prices and to reduce their payment outlays, subscribers show two reactions. For example, under self-elasticity, the subscribers remain insensitive to prices due to their consumption behavior. For example, the lighting mechanism is active during the evening hours, when the subscribers will least react to price changes in energy prices. However, in mutual elasticity, the subscribers will change the consumption hours. Therefore, the consumption is postponed by consumers to future or past periods. From here, a matrix can be created for load changes in relation to changes in the energy carrier's price, and the elasticity factors form the arrays of the matrix. Self-elasticity statements are always negative or zero because whenever energy carrier price goes up in an interval of time, energy demand decreases and cannot be relocated to other hours. Mutual elasticity lists are always positive or zero since whenever energy carrier price goes up, customers are forced to shift their consumption to other times [42,43]. The elasticity matrix is:

$$\begin{bmatrix} \frac{\partial P(1)}{P(1)} \\ \frac{\partial P(2)}{P(2)} \\ \vdots \\ \frac{\partial P(24)}{P(24)} \end{bmatrix} = \begin{bmatrix} E_t(1,1) & \cdots & E_t(1,24) \\ \vdots & \ddots & \vdots \\ E_t(24,1) & \cdots & E_t(24,24) \end{bmatrix} \begin{bmatrix} \frac{\partial \Lambda(1)}{\Lambda(1)} \\ \frac{\partial \Lambda(2)}{\Lambda(2)} \\ \vdots \\ \frac{\partial \Lambda(24)}{\Lambda(24)} \end{bmatrix} \quad (48)$$

Within this elasticity matrix, two different types of consumer response are represented. The main diagonal reflects how demand changes within the same time interval when price varies, while the remaining entries describe how consumption is redistributed across different intervals. When the off-diagonal values disappear, consumers respond to price increases mainly by reducing or shedding load rather than moving it to another period. In contrast, when the diagonal terms are negligible, the dominant response is load shifting, meaning that users adjust the timing of consumption to control cost. Moreover, nonzero values in the upper triangular part indicate a tendency to advance electricity use before expensive periods begin. By comparison, zero values in the lower triangular part imply that consumers delay demand until high-price intervals have passed, thereby avoiding exposure to costly periods [23,25].

A2.2 Single-period IBDR model (interruptible loads) based on linear function

For interruptible loads such as lighting, demand variation within a given time interval is governed only by the electricity price of that same interval, and consumption in one period does not carry over to others. In practical terms, when a subscriber curtails part of the lighting load during a peak-price period, that deferred demand cannot be shifted and recovered in another interval [44]. For this reason, this category of load is appropriately represented using a single-period model. The subscriber load size in group n ranges from i to j at time t due to the price change, such that:

$$P_{n,t}^{IBDR} = P_{n,t} + \Delta P_{n,t}^{IBDR} \quad (49)$$

where $\Delta P_{n,t}^{IBDR}$ is the amount of change in the subscriber consumption pattern compared to the case without DR. $P_{n,t}$ is the subscribers' power. Due to the change in the consumption pattern, there is a profit for subscribers, which is desirable to maximize. Therefore, the amount of profit for subscribers in exchange for participating in DR is equal to:

$$\delta_{n,t}^{IBDR} = \omega_{n,t}^{IBDR} - \left(P_{n,t}^{IBDR} \cdot \Lambda_{n,t}^{spot} \right) \quad (50)$$

where $\delta_{n,t}^{IBDR}$ is the subscriber's profit, $\omega_{n,t}^{IBDR}$ is the value of electricity from the customer's perspective, $P_{n,t}^{IBDR}$ is the power consumption of interruptible loads, and $\Lambda_{n,t}^{spot}$ is the current price of electricity for subscribers. To maximize the customer's profit, the derivative of the profit function with respect to the load must be zero. Therefore, considering the derivative of the profit function and using Taylor expansion, the subscriber's income function is equal to:

$$\frac{\partial \delta_{n,t}^{IBDR}}{\partial P_{n,t}^{IBDR}} = 0 \Rightarrow \frac{\partial \omega_{n,t}^{IBDR}}{\partial P_{n,t}^{IBDR}} = \Lambda_{n,t}^{spot} \Rightarrow \left\{ \begin{aligned} \omega_{n,t}^{IBDR} &= \omega_{n,t}^0 + \frac{\partial \omega_{n,t}^{IBDR}}{\partial P_{n,t}^{IBDR}} \Delta P_{n,t}^{IBDR} + \frac{1}{2} \frac{\partial^2 \omega_{n,t}^{IBDR}}{(\partial P_{n,t}^{IBDR})^2} (\Delta P_{n,t}^{IBDR})^2 \end{aligned} \right. \quad (51)$$

The initial conditions for subscriber profits in order to achieve maximum joint profit are equal to:

$$\delta_{n,t}^0 = \omega_{n,t}^0 - \left(P_{n,t}^{IBDR} \cdot \Lambda_{n,t}^0 \right) \Rightarrow \left\{ \begin{aligned} \frac{\partial \delta_{n,t}^0}{\partial P_{n,t}^{IBDR}} &= 0 \Rightarrow \frac{\partial \omega_{n,t}^0}{\partial P_{n,t}^{IBDR}} = \Lambda_{n,t}^0 \\ \frac{\partial^2 \omega_{n,t}^0}{(\partial P_{n,t}^{IBDR})^2} &= \frac{\partial \Lambda_{n,t}}{\partial P_{n,t}^{IBDR}} = \frac{1}{E_{i,i}} \cdot \frac{\Lambda_{n,t}^0}{P_{n,t}} \end{aligned} \right. \quad (52)$$

where $\delta_{n,t}^0$ is the basic profit of subscribers in the DR program, $\omega_{n,t}^0$ is the basic value of power from the subscribers' perspective, and $\Lambda_{n,t}^0$ is the basic price of the energy carrier. By substituting Eq. (51) into (52) and simplifying the relations, the quadratic function of the income of subscribers of group n resulting from participation in the interruptible load program is expressed as follows:

$$\omega_{n,t}^{IBDR} = \omega_{n,t}^0 + \Lambda_{n,t}^0 \Delta P_{n,t}^{IBDR} + \frac{\Lambda_{n,t}^0}{2E_{i,i}P_{n,t}} (\Delta P_{n,t}^{IBDR})^2 \quad (53)$$

Accordingly, the post-participation consumption function for subscribers in group n , considering the contribution of interruptible loads in incentive-based demand response programs, can be expressed by the following equation:

$$\text{if } \left\{ \begin{aligned} \Lambda_{n,t}^{spot} &= \Lambda_{n,t}^0 \left[1 + \frac{P_{n,t}^{IBDR} - P_{n,t}}{E_{i,i}P_{n,t}} \right] \\ \Lambda_{n,t}^{spot} - \Lambda_{n,t}^0 &= \Lambda_{n,t}^0 \frac{P_{n,t}^{IBDR} - P_{n,t}}{E_{i,i}P_{n,t}} \end{aligned} \right. \Rightarrow P_{n,t}^{IBDR} = P_{n,t} \left[1 + E_{i,i}^j \cdot \frac{\Lambda_{n,t}^{spot} - \Lambda_{n,t}^0}{\Lambda_{n,t}^0} \right] \quad (54)$$

A2.3 Multi-period IBDR (shiftable loads) model based on linear function

Unlike interruptible loads, shiftable loads allow consumers to lower electricity expenses by rescheduling their use away from peak-price periods [23,25]. In this case, the consumption decision in one interval is influenced not only by the local price but also by price levels in other intervals, since demand can be relocated across time. To capture this interdependence, this category of load is modeled through a multi-period formulation in which mutual elasticity is used to represent cross-period consumption adjustments.

$$E_{ij} = \frac{\Lambda_j^0 \partial P_i}{P_i^0 \partial \Lambda_j} \quad (55)$$

Accordingly, by applying the above definition and following the same procedure used for the single-period formulation, the governing equation for shiftable responsive loads can be derived as follows:

$$P_{n,t}^{IBDR} = P_{n,t} \cdot \left[1 + \sum E_{ij} \cdot \frac{\Lambda_{n,t}^{spot} - \Lambda_{n,t}^0}{\Lambda_{n,t}^0} \right] \quad (56)$$

A2.4 Complete model of responsive loads based on linear function

In practice, subscribers respond to price changes through both demand response pathways while aiming to minimize their overall electricity expenditure. For this reason, the complete responsive load formulation integrates the single-period representation with the multi-period framework into a unified model. That is to say, assuming that the proportion of subscribers belonging to group n of DR programs can be written as δ_p^{IBDR} , it is expressed as: (Eq. (57) expresses Eq. (19) in Section 2.4.)

$$P_{n,t}^{IBDR} = \delta_p^{IBDR} \cdot P_{n,t}^0 \cdot \left(\left[\Delta P_{n,t}^{IBDR} \cdot (1 + E_{ii}) \cdot \left(\frac{\Lambda_{n,t}^{spot} - \Lambda_{n,t}^0}{\Lambda_{n,t}^0} \right) \right] + \left[\sum_{j=1}^{24} \Delta P_{n,t}^{IBDR} \cdot (1 + E_{ij}) \cdot \left(\frac{\Lambda_{n,t}^{spot} - \Lambda_{n,t}^0}{\Lambda_{n,t}^0} \right) \right] \right) \quad (57)$$

Data availability

Data will be made available on request.

References

- [1] Fuzhong Yu, Shaoping Bi, Enming Zhang, Optimizing electric vehicle-based renewable energy microgrids with V2G technology: an integrated approach for grid efficiency and sustainability, *J. Energy Storage* 141 (2026) 119156.
- [2] Wenjun Liao, et al., Intelligent energy management: sizing and locating microgrids with hybrid electric vehicle and supercapacitor-battery hybrid energy storage, *J. Energy Storage* 139 (2025) 118849.
- [3] Mohammad Esmail Hassanzadeh, et al., Hierarchical optimal allocation of BESS using APT-FPSO based on stochastic programming model considering voltage sensitivity and eigenvalues analyses, *Int. J. Electr. Power Energy Syst.* 153 (2023) 109291.
- [4] Reza Sepehrzad, et al., Experimental and developed DC microgrid energy management integrated with battery energy storage based on multiple dynamic matrix model predictive control, *J. Energy Storage* 74 (2023) 109282.
- [5] Reza Sepehrzad, et al., Intelligent hierarchical energy and power management to control the voltage and frequency of micro-grids based on power uncertainties and communication latency, *Electr. Power Syst. Res.* 202 (2022) 107567.
- [6] Cong Zhang, et al., Vehicle-to-grid based optimization for managing electric vehicle charging load and reducing grid fluctuations in microgrids, *J. Energy Storage* 139 (2025) 118715.
- [7] Dan Liu, et al., Fixed-time quasi-consensus energy management method for battery energy storage systems in DC microgrids under two types of DoS attacks, *J. Energy Storage* 113 (2025) 115574.
- [8] Reza Sepehrzad, et al., A multi-agent deep reinforcement learning paradigm to improve the robustness and resilience of grid connected electric vehicle charging stations against the destructive effects of cyber-attacks, *Energy* 307 (2024) 132669.
- [9] Tareq Aziz, et al., A methodology to prevent cascading contingencies using BESS in a renewable integrated microgrid, *Int. J. Electr. Power Energy Syst.* 110 (2019) 737–746.
- [10] Hwei-Ming Chung, et al., Local cyber-physical attack for masking line outage and topology attack in smart grid, *IEEE Trans. Smart Grid* 10 (4) (2018) 4577–4588.
- [11] O.T. Thi Kim, T.H.T. Le, M.J. Shin, V. Nguyen, Z. Han, C.S. Hong, Distributed auction-based incentive mechanism for energy trading between electric vehicles and mobile charging stations, *IEEE Access* 10 (2022) 56331–56347.
- [12] D.T. Rizi, M.H. Nazari, M. Fani, S.H. Hosseini, Assessment of cyber security in renewable electricity market considering system reliability using machine learning, in: 2023 8th International Conference on Technology and Energy Management (ICTEM), Mazandaran, Babol, Iran, Islamic Republic of, 2023, pp. 1–5.
- [13] Charalampos Avraam, Luis Ceferino, Yuri Dvorkin, Operational and economy-wide impacts of compound cyber-attacks and extreme weather events on electric power networks, *Appl. Energy* 349 (2023) 121577.
- [14] Kevin A. Melendez, Yolanda Matamala, Adversarial attacks in demand-side electricity markets, *Appl. Energy* 377 (2025) 124615.
- [15] X.-K. Liu, S.-Q. Wang, M. Chi, Z.-W. Liu, Y.-W. Wang, Resilient secondary control and stability analysis for DC microgrids under mixed cyber attacks, *IEEE Trans. Ind. Electron.* 71 (2) (Feb. 2024) 1938–1947.
- [16] Daogui Tang, Yi-Ping Fang, Enrico Zio, Vulnerability analysis of demand-response with renewable energy integration in smart grids to cyberattacks and online detection methods, *Reliab. Eng. Syst. Saf.* 235 (2023) 109212.
- [17] Süleyman Yıldız, Burak Yildirim, Mahmut Temel Özdemir, Enhancing load frequency control and cybersecurity in renewable energy microgrids: a fuel cell-based solution with non-integer control under cyber-attack, *Int. J. Hydrog. Energy* 75 (2024) 438–449.
- [18] Yang Li, et al., Enhancing cyber-resilience in integrated energy system scheduling with demand response using deep reinforcement learning, *Appl. Energy* 379 (2025) 124831.
- [19] Amin Khodadadi, et al., Data-driven hierarchical energy management in multi-integrated energy systems considering integrated demand response programs and energy storage system participation based on MADRL approach, *Sustain. Cities Soc.* 103 (2024) 105264.
- [20] Rong Huang, Yuancheng Li, Xiao Wang, Attention-aware deep reinforcement learning for detecting false data injection attacks in smart grids, *Int. J. Electr. Power Energy Syst.* 147 (2023) 108815.
- [21] Hossein Mohammadi Rouzbahani, Hadis Karimipour, Lei Lei, Multi-layer defense algorithm against deep reinforcement learning-based intruders in smart grids, *Int. J. Electr. Power Energy Syst.* 146 (2023) 108798.
- [22] R. Sepehrzad, M.J. Faraji, A. Al-Durra, M.S. Sadabadi, "Enhancing Cyber-Resilience in Electric Vehicle Charging Stations: A Multi-Agent Deep Reinforcement Learning Approach," *IEEE Transactions on Intelligent Transportation Systems* Vol. 25, No. 11, Nov. 2024, pp. 18049–18062.
- [23] Jiandong Duan, et al., Optimal scheduling strategy with integrated demand response based on stepped incentive mechanism for integrated electricity-gas energy system, *Energy* 313 (2024) 133689.
- [24] Hamid Karimi, Ehsan Heydarian-Forushani, A multi-objective bi-level framework to model distribution system operator's behavior in the wholesale and local transactive markets, *Energy* 312 (2024) 133695.
- [25] Reza Sepehrzad, Amir Saman Godazi Langeroudi, Ahmed Al-Durra, Amjad Anvari-Moghaddam Sadabadi, Mahdieh S. Sepehrzad, Demand response-based multi-layer peer-to-peer energy trading strategy for renewable-powered microgrids with electric vehicles, *Energy* 320 (2025) 135206. ISSN 0360-5442.
- [26] O. Sener, V. Koltun, Multi-task learning as multi-objective optimization, *Adv. Neural Inf. Proces. Syst.* 31 (2018).
- [27] Fuzhang Wu, et al., Uncertain scheduling potential of charging stations under multi-attribute uncertain charging decisions of electric vehicles, *Appl. Energy* 374 (2024) 124036.
- [28] Peixiao Fan, et al., A multi-layer intelligent control strategy for multi-regional power system with electric vehicles: a deep reinforcement learning approach, *J. Energy Storage* 103 (2024) 114381.
- [29] C. Yang, Z. Zhou, H. Zhuang, C. Wang, M. Yang, Global pose initialization based on cridded gaussian distribution with wasserstein distance, *IEEE Trans. Intell. Transp. Syst.* 24 (5) (2023) 5094–5104.
- [30] Ziyun Yuan, et al., Soft sensor development for mixed oil interface tracking in multi-product pipelines based on knowledge-informed semi-supervised Variational Bayesian Gaussian mixture regression, *Energy* 300 (2024) 131516.
- [31] Md Shadman Abid, et al., A novel multi-objective optimization based multi-agent deep reinforcement learning approach for microgrid resources planning, *Appl. Energy* 353 (2024) 122029.

- [32] Jie Zhou, Rong Lu, Microgrid cooperative scheduling optimization based on quality of service constraint and deep Q-network algorithm, *Int. J. Low-Carbon Technol.* 20 (2025) 47–57.
- [33] Abdelkader Messlem, et al., Optimizing microgrid energy management using reinforcement double deep Q-networks with prioritized experience replay, in: 2024 3rd International Conference on Advanced Electrical Engineering (ICAEE), IEEE, 2024.
- [34] J. Liu, Z. Wang, C. Chen, D. Dong, Efficient Bayesian policy reuse with a scalable observation model in deep reinforcement learning, *IEEE Trans. Neural Networks Learn. Syst.* 35 (10) (Oct. 2024) 14797–14809.
- [35] X.I.A. Yi, et al., Scheduling optimization of master-slave game in microgrid based on Bayesian improved deep learning algorithm under dual carbon objective, *J. Electr. Eng.* 19 (2) (2024) 249–258.
- [36] Reza Sepehrzad, et al., Enhanced cyber-resilience in flexible energy markets for microgrids: a trust-aware federated deep reinforcement learning framework, *Appl. Energy* 406 (2026) 127282.
- [37] Reza Sepehrzad, et al., Secure FDRL for safe decision-making and policy embedding validation in hybrid microgrids under hybrid cyberattacks, *Int. J. Electr. Power Energy Syst.* 176 (2026) 111646.
- [38] Reza Sepehrzad, et al., Microgrid secure pricing and real-time flexible energy market integrated with trust-based demand response: a multi-layer FDRL architecture, *Int. J. Electr. Power Energy Syst.* 174 (2026) 111536.
- [39] Atefeh Hedayatnia, et al., Two-stage data-driven optimal energy management and dynamic real-time operation in networked microgrid based on a deep reinforcement learning approach, *Int. J. Electr. Power Energy Syst.* 160 (2024) 110142.
- [40] Sepehrzad Reza, et al., Multilevel energy management and optimal control system in smart cities based on deep machine learning, in: *Applications of deep machine learning in future energy systems*, Elsevier, 2024, pp. 265–314.
- [41] Hamid Reza Abdolmohammadi, Reza Sepehrzad, Ehsan Azad-Farsani, Adel Oubelaid, Nima Khosravi, A new gradient projection active set approach for reliable and economic operation of combined energy systems, *Energy Conversion and Management: X* 29 (2026) 101518 (ISSN 2590-1745).
- [42] Reza Sepehrzad, Ahmed Al-Durra, Amjad Anvari-Moghaddam, Mahdiah S. Sadabadi, Short-term and probability scenario-oriented energy management of integrated energy distribution systems with considering energy market interactions and end-user participation, *Energy* 322 (2025) 135691 (ISSN 0360-5442).
- [43] Reza Sepehrzad, Milad Moafi, Ahmed al Durra, Mahdiah S. Sadabadi, Supervisory-learning microgrid control strategy in the presence of energy storage systems based on uncertainties and energy market interactions to improve stability, *Energy* 342 (2026) 139738 (ISSN 0360-5442).
- [44] Reza Sepehrzad, et al., A critical overview of local energy communities: State-of-the-art, real-life applications & challenges and tackling the academia-industry gap, *Renew. Sust. Energ. Rev.* 226 (2026) 116165.